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Flame dynamics and combustion noise: progress and challenges

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ABSTRACT

This article proposes a review of the state of knowledge in the field of combustion noise. The survey comprises an initial discussion of indirect and direct noise sources and their general characteristics, a summary of expressions devised to estimate combustion noise from turbulent flames, a discussion of the fundamental equations describing sound emission from a reactive region and an evaluation of scaling methods for combustion noise. An account is provided of a set of experiments on noise radiation from perturbed laminar flames. Sources of intense radiation of sound are identified and theoretical expressions of the pressure field are compared with detailed measurements from well controlled experiments. These experiments indicate that flame dynamics determine to a great extent the radiation of sound from flames. This is further demonstrated with experiments dealing with effects of confinement. Links between combustion noise and combustion instabilities are drawn on this basis. These two aspects are usually treated separately but they are manifestations of similar processes. Much of the current effort in the field of combustion noise focuses on numerical estimation techniques using modern computational tools. The state of the art is less advanced than in computational aeroacoustics (CAA) but it is possible to foresee that computational combustion acoustics (CCA) will progressively evolve into a well established scientific field.

Key words: Combustion noise, Flame dynamics, Entropy noise

1. INTRODUCTION

It is now commonly admitted that combustion noise originates from two main sources. The first, designated as "indirect noise" is produced when temperature fluctuations due to the combustion process traverse an inhomogeneous region characterized by mean flow gradients. Indirect noise also designated as entropy noise (Candel and Marble (1977)) is generated by the coupling between a convective entropy mode and the mean

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flow velocity or pressure gradients. The second source, "direct noise", results from the nonsteady processes of volumetric expansion in the reactive region (Bragg (1963); Smith and Kilham (1963); Thomas and Williams (1966); Hurle et al. (1968)). The main source of direct noise results from the rate of change of heat release rate associated with the chemical conversion of the reactants (Strahle (1971, 1972, 1973); Dowling and Ffwocs Williams (1983)). The first source of noise is of importance in high performance propulsion systems because the combustor operates at elevated pressures and burnt gases are exhausted through turbine bladerows or directly through a nozzle. The coupling between hot spots generated by combustion and mean flow gradients then gives rise to pressure perturbations. The amplitude level of the radiated pressure depends in this case on the combustion mode and on the type and intensity of temperature fluctuations reaching the combustor exhaust section and convected into the mean flow velocity gradient. The second source depends more directly on the combustion process and is controlled by the nonsteady heat release taking place in the combustion region. This second process is also influenced by interactions with the boundaries, which partially reflect sound waves generated by combustion. Under normal operating conditions the noise sources are incoherent and radiate a broadband of frequencies ("combustion roar" see Mahan and Karchmer (1991)). Under unstable operation a feedback is established between the sound sources and pressure perturbations propagating in the system. Noise radiation becomes more coherent as it is tuned on one of the resonance frequencies of the system (see Candel (1992); Culick and Yang (1995); Candel (2002); Lieuwen and Yang (2005) for reviews of combustion dynamics). This usually involves a cyclic perturbation of the flow inducing a periodic motion of the flame and associated fluctuations in heat release rate. Unstable combustion gives rise to large amplitude sound pressure levels which are often detrimental to the operation of the system. The heat fluxes to the combustor walls are intensified, structural vibration is induced and this may lead in extreme cases to failure. Studies of combustion noise generally focus on situations where the flow dynamics can be considered to be independent of the radiated sound. It is implicitly assumed that the flow dynamics is decoupled from the induced wave motion and the sound emission from unstable flames is generally not considered when dealing with combustion noise. It is however not always possible to obtain a complete decoupling between the flow and its acoustic radiation because practical systems are confined and the boundaries reflect sound towards the reactive region. One scientific challenge in the analysis of acoustically coupled combustion is to estimate the level of oscillation when the unstable system has reached a limit cycle. A possible solution to this fundamental problem proposed by Noiray et al. (2008) consists in making use of the flame describing function to represent the combustion response to incoming perturbations. It is then possible to solve the system dispersion relation to obtain the modal growth rates as a function of amplitude and predict the level of oscillation corresponding to the limit cycle.

In all cases the sources of combustion noise (indirect and direct) are related to the flame dynamics and to unsteady rates of heat release. The analysis of combustion noise is then intimately tied to an understanding of dynamics of perturbed flames. This relationship has been explored in a set of recent studies. On the experimental side much

has been learnt from detailed experiments and more specifically from studies on laminar flames. These investigations indicate that intense sound emission originates from flame interactions. On the theoretical side, much of the work has relied on classical expressions derived by Strahle. The pressure field is expressed in terms of an integral of the time delayed rate of change of the nonsteady heat release rate. This expression can be used to derive scaling rules or develop statistical estimates of combustion noise. Alternatively it is possible to follow methodologies developed in computational aeroacoustics (CAA). One idea consists in calculating the reacting flow in an LES framework and determining the sound field from Strahle's result. Another approach uses an LES calculation to determine the sound sources which are then inserted as input to a perturbed set of Euler equations. These equations are then integrated to obtain the sound field. Another possibility consists in calculating the reactive flow and the sound field with the same code. There are still many difficulties to overcome to get clean and reliable estimates with these methods. One aspect which is still not settled is the identification of the noise source terms. In some calculations of noise from nonpremixed flames the source term is expressed as a function of the second time derivative of the density while other calculations use alternative source expressions. Indirect noise is also being investigated with modern computational tools for wave propagation in nonuniform flow allowing direct estimations of the sound radiation associated with this process. This article contains a review of progress in these various areas and a discussion of current challenges. It focuses on direct combustion noise and begins with an account of the classical theoretical developments in this area (section 2). Equations of combustion acoustics are discussed in section 3 to sort out the noise source terms. Standard expressions of the sound field radiated by open flames are used to develop scaling rules which are briefly discussed in section 4. Direct noise sources are examined by making use of systematic experiments on perturbed flames (section 5). Five types of flame geometries are investigated and the radiated sound field is linked to changes in flame surface area and to rates of change of heat release in these flames. Confinement effects are briefly considered in section 6. It is indicated in particular that confinement changes the acoustic response of the system to sources of noise but also modifies the flame dynamics with a direct impact on the associated nonsteady heat release source terms. The link between noise and instabilities is also discussed when positive feedback between noise and sources leads to resonant regimes. The final section contains a short survey of recent developments in computational combustion acoustics.

2. CLASSICAL COMBUSTION NOISE THEORY

2.1. Early investigations

A classical investigation of noise generated by combustion was carried out by Thomas and Williams (1966). The principle of this clever experiment was to fill soap bubbles with a reactive mixture and record the pressure field radiated by the burning of each isolated bubble. It was found that the pressure signal could be described as the sound generated by a monopole source of strength $d\Delta V/dt$, where ΔV represented the volume increase due to thermal expansion of the gases crossing the reactive front. In this early

model, the farfield pressure signal p' was expressed in terms of the volume acceleration induced by nonsteady combustion:

$$p'(\boldsymbol{r},t) = \frac{\rho_\infty}{4\pi r}\frac{d^2\Delta V}{dt^2} \tag{1}$$

where ρ_∞ is the farfield air density at the measurement location $\boldsymbol{r}$ and r designates the distance between the compact flame and the observation point. According to eq. (1) combustion noise results from the flow expansion determined by the heat release rate. It was argued that because this source is isotropic there is no preferential direction of radiation. These experiments carried out in the laminar case strongly support some earlier theoretical investigations of turbulent combustion noise. In the theory established by Bragg (1963), it was postulated that a turbulent flame behaves as a net monopole radiator. An experimental validation of this theory was carried out by Hurle et al. (1968). Starting from eq. (1) and assuming that a turbulent flame is acoustically equivalent to a distribution of monopole sources of sound with different strengths and frequencies distributed throughout the reaction zone (Smith and Kilham (1963)), it was possible to express the farfield sound pressure p' radiated by the turbulent reacting flow as a function of the volumetric rate of consumption of reactants q by the flame (Hurle et al. (1968); Price et al. (1968)):

$$p'(\boldsymbol{r},t) = \frac{\rho_\infty}{4\pi r}\left(\frac{\rho_u}{\rho_b}-1\right)\left[\frac{dq}{dt}\right]_{t-r} \tag{2}$$

In this expression, $\rho_u/\rho_b = V_b/V_u$ designates the volumetric expansion ratio of burnt to unburnt gases, and τ is the time required for acoustic propagation from the combustion region to the measurement point r. In eq. (2), it is assumed that acoustic wavelengths λ are large compared to any of the characteristic scales of the flow ($\lambda \gg L$) and that the measurement point r is far from the source region ($r \gg \lambda$). The previous expression provides the sound pressure in the farfield for a compact source region, but it can be used indifferently for premixed or nonpremixed flames (Price et al. (1968)).

In the premixed case and for lean conditions (equivalence ratio less than one) the volumetric rate of reactants consumption q can be estimated from the light emission intensity I of excited radicals like C_2^* or CH* (Hurle et al. (1968); Shivashankara et al. (1975)) and OH^* (Katsuki et al. (1986)) in the reaction zone. This yields an elegant optical method to measure the volumetric rate of reactants consumption:

$$q = kI \tag{3}$$

The coefficient k depends on the fuel, the free radical observed, the combustion regime, the flame shape and type and the experimental setup. This relation can be used to study combustion noise from a homogeneous reactive mixture submitted to flow perturbations. Concerning inhomogeneous mixtures, the relation between the radiated pressure and the flame chemiluminescence is not as straightforward as in the premixed

case and one cannot deduce the rate of reaction from emission from excited free radicals. Considering only premixed flames the linearity coefficient k is determined from separate experiments by plotting the mean emission intensity $\overline{I}$ versus the mean flowrate all other parameters remaining fixed. Combining equations (2) and (3), the radiated farfield sound pressure field can be linked to the light emitted from the turbulent combustion region:

$$p'(\boldsymbol{r},t) = \frac{\rho_\infty}{4\pi r}\left(\frac{\rho_u}{\rho_b} - 1\right) k \left[\frac{dI}{dt}\right]_{t-\tau} \tag{4}$$

This expression was checked in many early experiments on premixed flames held at a fixed equivalence ratio and light intensity emitted by selected free radicals has been used extensively to get an estimate of the heat release rate (Hurle et al. (1968)). Assuming that fuel and oxidizer react in stoichiometric proportions, it was also successfully applied to turbulent diffusion flames by keeping a constant global mixture ratio (Price et al. (1968)).

These initial developments led to the more complete combustion noise theory presented in what follows.

2.2. Classical pressure field expressions

Efforts to clarify and improve Bragg's theory were undertaken by Strahle (1971, 1972). Using the balance equations of fluid dynamics and Lighthill's analogy for aerodynamic sound (Lighthill (1952)), it is possible to derive an expression for the sound radiated from a region undergoing turbulent combustion. This can be used to link farfield acoustic pressure fluctuations to density fluctuations inside the combustion region (Strahle (1971)):

$$p'(r,t) = c_0^2 \rho'(r,t) = -\frac{1}{4\pi r}\frac{\partial^2}{\partial t^2}\int_V \rho'_T\left(\boldsymbol{r}_0, t - \frac{|\boldsymbol{r}-\boldsymbol{r}_0|}{c_0}\right) dV(\boldsymbol{r}_0) \tag{5}$$

In this expression, ρ'_T are density fluctuations dominated by turbulent fluctuations and not by the acoustic field. According to Strahle (1971), combustion noise is generated by the violent density changes inside the combustion region. Equation (5) also accounts for interferences at the measurement location between the different acoustical sources distributed inside the reacting zone which radiate with different delayed time delays $t^* = t - |\boldsymbol{r} - \boldsymbol{r}_0|/c_0$. The relation of the sound output to the reaction rate is established in second article (Strahle (1972)). In modern approach of combustion noise, this relation can be obtained using Green's theorem (see for example Dowling and Ffwocs Williams (1983)). It was first derived starting from eq. (5) and a mass balance over the combustion region that yields an expression for the farfield sound pressure as a surface integral:

$$p'(r,t) = \frac{\overline{\rho}_1}{4\pi r}\int_{S_1} \frac{\partial \boldsymbol{v}_t}{\partial t}\left(\boldsymbol{r}_0, t - \frac{|\boldsymbol{r}-\boldsymbol{r}_0|}{c_0}\right) \cdot \boldsymbol{n}_0 \, dS(\boldsymbol{r}_0) \tag{6}$$

where v_t correspond to turbulent velocity fluctuations induced by strong fluctuations of density inside the combustion zone, S_1 is a surface which encloses the outer flame surface in the burnt gases region, $\boldsymbol{n}_0$ is the outward normal direction taken on the surface S_1 and $\overline{\rho}_1$ is the mean density on the surface S_1 (burnt gases). An important conclusion is that the normal velocity fluctuations on the outer flame surface, produced by strong interior density fluctuations inside the flame induce significant sound output. Combined with a balance of enthalpy, the integral over the outer flame surface S_1 can be replaced by an integration over the volume V_c containing the combustion region. This finally yields:

$$p'(r,t) = \frac{\gamma - 1}{4\pi r c_1^2} \int_{V_c} \frac{\partial \dot{Q}}{\partial t} \left(\boldsymbol{r}_0, t - \frac{|\boldsymbol{r} - \boldsymbol{r}_0|}{c_0} \right) dV\,(\boldsymbol{r}_0) \tag{7}$$

In this expression, γ is the specific heat ratio and c_1 is the sound speed in the burnt gases and $\dot{Q}$ the volumetric heat release rate in the combustion zone. The domain of validity of eq. (5) and eq. (6) is restricted to low Mach numbers ($M \ll 1$), to the farfield ($|\boldsymbol{r} - \boldsymbol{r}_0| \gg \lambda$), and to situations where the acoustic wavelength greatly exceeds the typical dimensions of the flame ($\lambda \gg L$). The last expression originally obtained by Strahle can be derived using various other approaches presented later in this study. It constitutes the main result of combustion noise theory. The farfield pressure radiated from a turbulent flame results from the contribution over the combustion region of elementary sources formed by the local time rates of change of the heat release rate. Each source produces an elementary contribution at the measurement location after a time delay corresponding to propagation between the source inside the combustion region and the point of observation in the farfield. Expression (7) proves the relation (4) obtained on a more intuitive basis by Hurle et al. (1968) and is equally applicable to premixed or non-premixed flames.

This brief review indicates that sound radiation from flames results from the unsteadiness of the combustion process. Intense levels of noise are produced by interactions involving rapid changes of the rate of chemical conversion. When the flame zone is compact with respect to the acoustic wavelength the time delays between the different sources of sound can be neglected and noise radiation is essentially isotropic. Some directivity can be due to propagation of sound through flow field inhomogeneities and specifically temperature gradients (Strahle (1973)), from source convection effects or from other sources of sound neglected in equation (7).

2.3. Directivity

While combustion noise is essentially radiated isotropically, experiments on open turbulent flames indicate that there are some directivity effects. Smith and Kilham (1963) found that the preferred directions of acoustic propagation for the flames tested lie between 40° and 80° from the burner axis. These authors attribute the measured directivity to convection of sound sources by the flow. Price et al. (1968) also confirms the quasi-isotropic character of noise radiation when $\lambda \gg L$ with a preferred direction at 80° for long turbulent non premixed jet flames when the flame length L is no longer

small compared to λ. On the theoretical side, sound directivity is often attributed to the finite combustion region size L relative to the wavelength λ (Strahle (1972)). Estimations of the contributions to the net sound output of the different source terms in the Lighthill's tensor (neglected in eq.(7)) are examined in references Strahle (1972, 1973); Hassan (1974); Doshiba and Hirano (1997). Hassan (1974) finds that the noise generated by combustion results from the superposition of a strong monopole, a weak dipole, and negligible quadrupole sources.

2.4. Premixed flames

Considering the case of premixed flames, it is noted by Thomas and Williams (1966) that sound radiation can be related to the rate of change of the flame surface area by assuming that the burning velocity is constant. Similarly Abugov and Obrezkov (1978) and later Clavin and Siggia (1991) indicate that in the wrinkled flame regime, the rate of chemical conversion is directly linked to the flame surface area $A(t)$ and derive the following expression for the radiated power:

$$W_a = \frac{\rho_\infty}{4\pi c_\infty}\left(\frac{1}{\rho_b} - \frac{1}{\rho_u}\right)^2 (\rho_u S_L)^2 \overline{\left(\frac{dA}{dt}\right)^2} \tag{8}$$

where S_L is the laminar burning velocity and the overline designates an averaging operator. It is difficult to test this expression in the general case of a turbulent flame because the instantaneous surface area is not directly accessible experimentally. Some validations are reported by Belliard (1997) and Truffaut (1998) for strongly turbulent flames. It is easier to use periodically perturbed laminar flame configurations to determine the instantaneous flame surface area and express the pressure signal to the rate of change of surface area. For a mixture of fresh reactants at a constant equivalence ratio, the pressure field is directly linked to the instantaneous flame surface area. Assuming a constant laminar burning velocity, one may extract the pressure signal from a direct measurement of the wrinkled flame surface area:

$$p'(\boldsymbol{r},t) = \frac{\rho_\infty}{4\pi r}\left(\frac{\rho_u}{\rho_b} - 1\right) S_L \left[\frac{dA}{dt}\right]_{t-\tau} \tag{9}$$

Alternatively, this can be accomplished by measuring light emission from free radicals in the flame, deducing from this measurement the heat release rate and determining the resulting pressure field from expression (4). Such direct comparisons are described in section 5 for a variety of flame configurations. At this point one should mention that the laminar burning velocity is not constant but depends on the local strain rate and curvature. It is shown theoretically that these effects act in combination in the form of flame stretch (Markstein (1964); Pelcé and Clavin (1982); Matalon and Matkowsky (1983)) and that this may induce some additional radiation of sound. Mechanisms inducing an acceleration of the rate of consumption during the terminal stage of the burning of small pockets of a reactive mixture may also lead to noise amplification as

proposed by Kidin et al. (1984). The following expression is thus probably applicable to a wider variety of problems:

$$p'(\boldsymbol{r},t) = \frac{\rho_\infty}{4\pi r}\left(\frac{\rho_u}{\rho_b} - 1\right)\left[\frac{d(S_d A)}{dt}\right]_{t-\tau} \tag{10}$$

where the laminar burning velocity was replaced by S_d that may also depend on stretch and time.

2.5. Nonpremixed flames

The sound field from non premixed and partially premixed flames is less well characterized than that radiated by premixed flames. It is however the subject of some recent experiments (Singh et al. (2005)) and of many numerical calculations (see for example Klein and Kok (1999)). Theoretical analysis is more difficult because the rate of heat release is less easily modeled in this case. This quantity also cannot be extracted from direct measurements of free radical emission.

2.6. Nearfield effects

The observation point is in the nearfield when the distance r between the source region and the detector is no longer large compared to the characteristic size L of the flame. Sound radiation in this region is less well documented but of importance in practical applications because the flame dimension can be quite large. Smith and Kilham (1963) find that for a distance $r < 30d$, where d is the burner diameter, the acoustic intensity ceases to vary like the inverse of the distance square $1/r^2$. Sound in this region is influenced by the closest noise sources. These nearfield effects are of primary importance for noise production in a confined environment. It will be shown later in this article that results obtained in the farfield region can be extrapolated to some extent to the nearfield. This can be used to analyse combustion instabilities coupled by noise generation in the flame and emission. An example is given by Noiray et al. (2006) where synchronized nearfield effects lead to a coherent amplification of the sound output from a distribution of identical flames. Many current studies of combustion noise yield nearfield results deduced from LES calculations in combination with acoustic perturbation equations. There is also an attempt by Lee and Lieuwen (2002) to determine the nearfield characteristics of perturbed conical flames based on a kinematic description.

3. EQUATIONS OF COMBUSTION ACOUSTICS

It is now worth examining the dynamic equations governing sound radiation from a region comprising N chemically reacting species. The starting point is a standard set of balance equations of mass, momentum, energy and species:

$$\frac{d\rho}{dt} = -\rho\boldsymbol{\nabla}\cdot\upsilon \tag{11}$$

$$\rho\frac{d\boldsymbol{v}}{dt} = -\nabla p + \nabla\cdot\tau \tag{12}$$

$$\rho c_p \frac{dT}{dt} = \dot{Q} + \frac{dp}{dt} + \boldsymbol{\tau} : \boldsymbol{\nabla \upsilon} - \boldsymbol{\nabla} \cdot \boldsymbol{J}^H \tag{13}$$

$$\rho \frac{dY_i}{dt} = \dot{\omega}_k - \boldsymbol{\nabla} \cdot \boldsymbol{J}_K^D \tag{14}$$

where $\dot{Q} = -\sum_{k=1}^{N} h_k \dot{\omega}_k = (-\Delta h_f^0)\dot{\omega}$ represents the heat release rate per unit volume, ρ, p, υ, Y_k, T and $\dot{\omega}_k$ respectively designate the local density, pressure, velocity vector, species mass fraction, temperature and mass production rate per unit volume of species i. In the previous set of equations τ, $\boldsymbol{J}^H$, and $\boldsymbol{J}_k^D$ respectively designate the viscous stress tensor, heat and mass diffusion fluxes. The mass diffusion flux $\boldsymbol{J}_k^D = \rho Y_k \boldsymbol{V}_k^D$ is represented in terms of species diffusion velocities $\boldsymbol{V}_k^D$ and the heat diffusion flux is given by the Fourier law and also accounts for heat transfer due to species diffusion $\boldsymbol{J}^H = -\lambda \Delta T + \sum_{k=1}^{N} \rho Y_k \boldsymbol{V}_k^D h_k$ where λ is the thermal conductivity. It is assumed that all species obey the perfect gas law. The state equation for the mixture is then given by

$$p = p r_g T, \text{ where } r_g = RW, \text{ and } \frac{1}{W} = \sum_{k=1}^{N} \frac{Y_k}{W_k} \tag{15}$$

In the previous expressions, R designates the perfect gas constant, r_g is the coefficient appearing in the mixture state equation which may vary during the reaction and the molar mass of the mixture W is given in terms of the molar masses of the various species W_k. The mixture enthalpy h and specific heat c_p are obtained by summing over the species: $h = \sum_{k=1}^{N} Y_k h_k$ and $c_p = \sum_{k=1}^{N} Y_k c_{pk}$.

To derive a wave equation from the previous system one may first express the temperature differential as a function of pressure, density and species differentials:

$$dT = \left(\frac{\partial T}{\partial p}\right)_{\rho, Y_k} dp + \left(\frac{\partial T}{\partial \rho}\right)_{\rho, Y_k} d\rho + \left(\frac{\partial T}{\partial Y_k}\right)_{\rho, p} dY_k \tag{16}$$

For a perfect gas, this expression becomes:

$$dT = \frac{T}{p} dp - \frac{T}{p} d\rho - T \sum_{k=1}^{N} \frac{W}{W_k} dY_k \tag{17}$$

Using this relation together with equations (13) and (14) one obtains after some rearrangements:

$$\begin{aligned} \frac{1}{\gamma p}\frac{dp}{dt} - \frac{1}{\rho}\frac{d\rho}{dt} = {} & \frac{\dot{Q}}{\rho c_p T} W \frac{d}{dt}\left(\frac{1}{W}\right) \\ & + \frac{1}{\rho c_p T}\left[\boldsymbol{\nabla} \cdot \lambda \boldsymbol{\nabla} T + \boldsymbol{\tau} : \boldsymbol{\nabla \upsilon} - \sum_{k=1}^{N} \rho Y_k c_{pk} \boldsymbol{V}_k^D \cdot \boldsymbol{\nabla} T\right] \end{aligned} \tag{18}$$

This expression may now be used in conjunction with equations (11) and (12) to derive a wave equation for the logarithm of the pressure. This is accomplished by dividing both equations by ρ, taking the material derivative of the first and subtracting the divergence of the second (see for example Candel et al. (1996)). After some rearrangements this process yields:

$$\begin{aligned}\boldsymbol{\nabla}\cdot\left(\frac{c^2}{\gamma}\boldsymbol{\nabla}\mathrm{In}\,p\right)-\frac{d}{dt}\left(\frac{1}{\gamma}\frac{d}{dt}\mathrm{In}\,p\right)&=\boldsymbol{\nabla}\cdot\left(\frac{1}{\rho}\nabla\cdot\boldsymbol{\tau}\right)\\&-\frac{d}{dt}\left(\frac{1}{\rho c_pT}\left[\boldsymbol{\nabla}\cdot\lambda\boldsymbol{\nabla}T+\boldsymbol{\tau}:\boldsymbol{\nabla}\boldsymbol{\upsilon}-\sum_{k=1}^{N}\rho Y_kc_{pk}\boldsymbol{V}_k^D\cdot\boldsymbol{\nabla}T\right]\right)\\&-\frac{d}{dt}\left(\frac{\dot{Q}}{\rho c_pT}\right)-\frac{d}{dt}\left[W\frac{d}{dt}\left(\frac{1}{W}\right)\right]-\boldsymbol{\nabla}\boldsymbol{\upsilon}:\boldsymbol{\nabla}\boldsymbol{\upsilon}\end{aligned}\tag{19}$$

where $\Delta\upsilon : \Delta\upsilon = (\partial\upsilon_i/\partial x_j)(\partial\upsilon_j/\partial x_i)$ was identified by Phillips (1960) to be the source of aerodynamic sound in non reactive turbulent flows. The left hand side of equation (19) has the form of a modified wave operator acting on the logarithm of the pressure while the right hand side can be considered to represent source terms of different nature. The separation between the left and right hand sides is somewhat arbitrary because some of the source terms also contain variables characterizing the wave field (Doak (1973); Kotake (1975); Goldstein (1976)). The separation adopted in eq. (19) is however reasonable and has become quite standard when the sound field only weakly interacts with the sources. It is generally admitted that all the terms associated with transfer of momentum, species and heat at the molecular level do not constitute important source terms and can be neglected altogether. One may then eliminate the source terms appearing on the first and second lines in eq. (19) to write:

$$\begin{aligned}\nabla\cdot\left(\frac{c^2}{\gamma}\boldsymbol{\nabla}\mathrm{In}\,p\right)-\frac{d}{dt}\left(\frac{1}{\gamma}\frac{d}{dt}\mathrm{In}\,p\right)&=\\-\frac{d}{dt}\left(\frac{\dot{Q}}{\rho c_pT}\right)-\frac{d}{dt}\left[W\frac{d}{dt}\left(\frac{1}{W}\right)\right]&-\nabla\upsilon:\nabla\upsilon\end{aligned}\tag{20}$$

Assuming that the right hand side terms in eq. (20) are known, the problem is equivalent to an inhomogeneous wave equation. This approach originally successfully applied by Lighthill (1952) for the study of aerodynamic sound produced by a turbulent non reactive flow is valid if the sources on the right hand side do not interact with the pressure field radiated. This is the case in a free field configuration or when interactions with boundaries are weak, but this separation is less well justified when the system becomes unstable and pressure waves couple with heat release rate perturbations.

Assuming for the moment free field conditions, i.e. sound is radiated from the source region without reflection from external boundaries. The third term appearing in the right hand side terms of eq. (20) is already identified as the source of aerodynamic sound. The first term is dominated by perturbations in the heat release rate $\dot{Q}$.

For the typical low Mach numbers and high Reynolds numbers (M << 1, Re >> 1) encountered in combustion, an order of magnitude analysis indicates that the dominant noise source is associated with perturbations of the heat release rate (Hassan (1974); Kotake (1975)). Velocity fluctuations intervene through the double scalar product of their gradient but are of lesser importance. Since combustion also induces changes in the molar composition, the molar mass of the mixture changes in the flame and this induces variations of the gas constant. This constitutes another source of noise which appears in the form as $d/dt(W[d(1/W)/dt])$ and can also be written as $d^2(\ln r_g)/dt^2$. The existence of an additional source of noise associated with non-isomolar combustion is also examined by Truffaut et al. (1998) who indicate that sound from flames may also originate from the fact that the number of moles changes when reactants are converted into products. These authors propose to write the corresponding source in the form:

$$\frac{d}{dt}\left(\frac{\dot{n}}{n}\right) \tag{21}$$

where $n = \rho/W$ is the number of moles per unit volume and $\dot{n}$ is the molar production rate per unit volume. It can be shown however that this term does not quite coincide with the source term found previously. Neglecting diffusion fluxes, noting that $n = \sum_{k=1}^{N} n_k = \rho\sum_{k=1}^{N} Y_k/W_k$ and using equation (14) of one finds that:

$$\frac{\dot{n}}{n} = \frac{1}{n}\frac{dn}{dt} = W\frac{d}{dt}\left(\frac{1}{W}\right) + \frac{1}{\rho}\frac{d\rho}{dt} \tag{22}$$

This expression contains an extra term $1/\rho(d\rho/dt)$ which is not substantiated. Under these circumstances it is preferable to use the valid form $d^2(\ln r_g)/dt^2$.

Equation (20) was derived without making any assumptions on the amplitude of the pressure field radiated by the reacting system. One may now consider that the pressure level is small compared to the local mean pressure $\ln p \simeq p'/\bar{p}$ and assume that the system is nearly isobaric so that $\bar{p} \simeq$ const. This is quite reasonable for most low Mach number combustion systems, see for example Williams (1985). Noting that $\gamma\bar{p} = \overline{\rho c}^2$ a first order perturbation analysis of eq. (20) yields for a low speed reactive flows where the convective term in the material derivative may be neglected $d/dt \sim \partial/\partial t$:

$$\nabla\cdot(\bar{c}^2\nabla p') - \frac{\partial^2 p'}{\partial t^2} = -\frac{\partial}{\partial t}\left[(\gamma-1)\dot{Q}'\right] - \gamma\bar{p}\nabla\boldsymbol{v}:\nabla\boldsymbol{v} - \frac{\gamma\bar{p}}{\bar{W}}\frac{\partial^2 W'}{\partial\tau^2} \tag{23}$$

In this expression, prime quantities correspond to fluctuations while overlined quantities designate mean values which may vary inside the combustion zone with the spatial coordinates. Mean quantities mainly depend on the local temperature and molar weight inside the combustion region. The mixture molar weight can be considered constant in most practical combustion systems when the oxidizer is air, because the mixture is then highly diluted with nitrogen. The last term influences sound production in oxygen enhanced combustion (Truffaut et al. (1998)) but can be neglected in most other applications

where the mean molar composition is nearly constant. The second term on the right hand side is the source of aerodynamic sound and is usually negligible in the relatively low speed flows prevailing in combustors.

The sound speed appearing in the spatial operator in the left hand side changes with the local temperature and this influences propagation from the sound sources to the observation point. Equation (23) is probably more adequate (Hedge et al. (1988); Laverdant and Thévenin (2003)) than a simpler version obtained by considering that $\overline{c}^2$ is constant and writing:

$$\nabla^2 p' - \frac{1}{\overline{c}^2}\frac{\partial^2 p'}{\partial t^2} = -\frac{1}{\overline{c}^2}\frac{\partial}{\partial t}\left[(\gamma-1)\dot{Q}'\right] - \overline{p}\nabla\boldsymbol{\upsilon} : \nabla\boldsymbol{\upsilon} - \frac{\rho}{\overline{W}}\frac{\partial^2 W'}{\partial t^2} \tag{24}$$

It is however much easier to use this last equation and derive expressions for the wave field by making use of Green's functions (Morse (1986)). Assuming that radiation takes place in an unconfined domain, only keeping the source term associated with perturbations in heat release and replacing $\overline{c}$ by c_0 one finds:

$$p'(\boldsymbol{r},t) = \frac{\gamma-1}{4\pi c_0^2}\int_V \frac{1}{|\boldsymbol{r}-\boldsymbol{r}_0|}\frac{\partial}{\partial t}\dot{Q}'\left(\boldsymbol{r}_0, t - \frac{|\boldsymbol{r}-\boldsymbol{r}_0|}{c_0}\right) dV(\boldsymbol{r}_0) \tag{25}$$

When the observation point is in the farfield and if the source region is compact, the previous expression becomes:

$$p'(\boldsymbol{r},t) = \frac{\gamma-1}{4\pi c_0^2 r}\int_V \dot{Q}'\left(\boldsymbol{r}_0, t - \frac{r}{c_0}\right) dV(\boldsymbol{r}_0) \tag{26}$$

This expression slightly differs from the original one (7) derived by Strahle that should be corrected by a factor ρ_0/ρ_1 to enclose all contributing noise sources (Crighton et al. (1992)) that were omitted in eq. (6). This point is also highlighted by Kotake and Takamoto (1990) in his study of the role of turbulence levels at the burner inlet in the sound field radiated by flames.

It is also interesting to note that contributions to indirect combustion or entropy noise do not appear explicitly in the preceding derivations. These noise sources are only relevant in gas turbine or jet engine. They are attributed to hot or cold spots that are created in the combustion chamber and persist. When these entropy perturbations are accelerated by the flow through the turbine nozzle they constitute additional noise sources (Strahle (1985)). The relative contribution to the overall radiated noise from indirect and direct combustion noise components is still an ongoing issue. It is shown hereby how to get an explicit expression of the noise sources using the same formalism developed above.

Considering that in most practical problems the flow speed is small enough in order to maintain combustion, density and temperature vary strongly with space and time within this region but density changes result primarily from thermal expansion and the

local motion resembles that of an incompressible flow with relatively small pressure fluctuations in the source region. This hypothesis is the base of the following developments. Equation (20) is now reinterpreted. It can be rewritten in the limit of low Mach flow $d/dt \sim \partial/\partial t$:

$$\nabla\cdot\left(\frac{1}{\rho}\nabla p\right)-\frac{\partial}{\partial t}\left(\frac{1}{\rho c^2}\frac{\partial}{\partial t}p\right)= -\frac{\partial}{\partial t}\left(\frac{\dot{Q}}{\rho c_p T}\right)-\frac{d}{dt}\left[W\frac{\partial}{\partial t}\left(\frac{1}{W}\right)\right]-\nabla\boldsymbol{v}:\nabla\boldsymbol{v} \tag{27}$$

In this expression, pressure logarithms appearing in the partial derivatives were developed and the identity $\gamma p = \rho c^2$ was used to simplify the left hand side operator. This expression can be rearranged formally adding on both sides of the equation the left hand side terms where the density and the sound speed are replaced by their uniform ambient values ρ_0 and c_0 as it is done for the analysis of aerodynamic sound (Lighthill (1952)):

$$\begin{aligned}\nabla^2 p-\frac{1}{c_0^2}\frac{\partial^2 p}{\partial t^2}= &\rho_0\frac{\partial}{\partial t}\left[\left(\frac{1}{\rho c^2}-\frac{1}{\rho_0 c_0^2}\right)\frac{\partial p}{\partial t}\right]-\rho_0\nabla\cdot\left[\left(\frac{1}{\rho}-\frac{1}{\rho_0}\right)\nabla p\right]\\ &-\rho_0\frac{\partial}{\partial t}\left(\frac{\dot{Q}}{\rho c_p T}\right)\\ &-\rho_0\nabla\mathbf{v}:\nabla\mathbf{v}-\rho_0\frac{\partial}{\partial t}\left[W\frac{\partial}{\partial t}\left(\frac{1}{W}\right)\right]\end{aligned} \tag{28}$$

This wave equation was also obtained by Howe (1998) in his analysis of combustion noise, although the last term involving composition fluctuations is missing in his derivation. The first two terms on the right hand side are responsible of indirect combustion noise emission as discussed by Howe (1998). The first term is a monopole whose strength is determined by differences between the adiabatic compressibility inside the source region and the ambient medium. As $\rho c^2 = \gamma p$ is quasi-constant through the combustion region, this noise source should be small. The second term has a dipole character resulting from differences between the acceleration of fluid of density ρ and that of a fluid of ambient medium density would experience under the same pressure gradient. As the density acts through its inverse $1/\rho$, the strength of this dipole is principally determined by the strength of the mean pressure gradient experienced by the flow. It can be ignored when regarding the noise radiated by open flames, but may be an important contribution inside combustors equipped with a nozzle. Convection of entropy inhomogeneties through a nozzle are known source of noise (Candel and Marble (1977)). This term is competed by direct combustion noise production resulting from heat release rate fluctuations. The last remaining terms in eq. (28) were already discussed. Direct combustion production is associated with the well known third term

which behaves as a monopole radiator. The following term is the classical Lighthill quadrupole associated with velocity fluctuations ($\rho_0\ \nabla\mathbf{v}:\nabla\mathbf{v} = (\partial^2/\partial x_i\,\partial x_j)(\rho_0 v_i v_j)$ because the velocity field was supposed divergence free inside the combustion region). The last source is a monopole that can generally be neglected in air burning systems.

While direct combustion noise largely prevails for farfield sound radiation from flames open to atmospheric conditions, relative contributions of direct to indirect noise remains unclear (Strahle (1985)). This aspect is also reinforced when looking at the near field sound production. While dipoles are less efficient radiator in the farfield compared to monopole sources because they exhibit some cancellation due to sources of opposite signs, the pressure field in the vicinity of the source scales with $1/r^2$ where r is the distance to the source and is therefore more intense than the nearfield sound produced by a monopole whose strength scales with $1/r$ (Dowling and Ffwocs Williams (1983)). As the sound field is limited by boundaries of the system relatively close to the sources, near field effects are important inside the combustor.

Now if the main principles underlying the description of the far-field noise radiated from a compact turbulent flame in free-field conditions are settled, nearfield effects, interactions with boundaries and the detailed mechanisms leading to intense noise production remain fairly unclear. These problems will be addressed in the following sections. Some flame flow interactions at the origin of intense combustion noise emission are studied in section 5. These mechanisms may be viewed as elementary interactions occurring in turbulent burners (Ducruix et al. (2003)).

4. COMBUSTION NOISE SCALING LAWS

Scaling laws can be derived from basic expressions for the pressure in the farfield in a manner which parallels that used by Lighthill in his classical analysis of jet noise. The radiated power is obtained for example by integrating the acoustic intensity over a large sphere. An estimate is required for the source correlation function. The scaling law then depends on the accuracy of the correlation function estimate. It is convenient to start from (26) and write the acoustic power as:

$$W_a = \frac{\overline{p'^2}}{\rho_0 c_0} 4\pi r^2 = \frac{(\gamma-1)^2}{4\pi\rho_0 c_0^5} \overline{\int_V \frac{\partial \dot{Q}'}{\partial t}(\boldsymbol{r}'_0,t)\frac{\partial \dot{Q}'}{\partial t}(\boldsymbol{r}''_0,t)d\boldsymbol{r}'_0 d\boldsymbol{r}''_0} \tag{29}$$

where the overline symbol designates an ensemble average. It is standard to use a change of integration variables $\boldsymbol{r}' = \boldsymbol{r}'_0$ and $\boldsymbol{r}''_0 - \boldsymbol{r}'_0 = \xi_0$ to write:

$$W_a = \frac{\overline{p'^2}}{\rho_0 c_0} 4\pi r^2 = \frac{(\gamma-1)^2}{4\pi\rho_0 c_0^5} \overline{\int_V \frac{\partial \dot{Q}'}{\partial t}(\boldsymbol{r}'_0,t)\frac{\partial \dot{Q}'}{\partial t}(\boldsymbol{r}'_0+\boldsymbol{\xi}_0,t)d\boldsymbol{r}'_0 d\boldsymbol{\xi}_0} \tag{30}$$

The correlation function term in this expression may now be estimated by considering that the time scale is defined by an inverse characteristic frequency f_c and that a correlation volume V_{cor} can be used to represent the spatial extent of heat release fluctuations (Hirsch et al. (2007)). This yields:

$$W_a = \frac{\overline{p'^2}}{\rho_0 c_0} 4\pi r^2 = \frac{(\gamma-1)^2}{4\pi\rho_0 c_0^5} f_c^2 \dot{Q}'^2_{max} V V_{cor} \tag{31}$$

It is then possible to estimate the maximum heat release fluctuation by considering that it is a fraction η of the volumetric heat release in the system: $\dot{Q}'_{max} = \eta \dot{m}_F h/V$ where $\dot{m}_F$ is the fuel mass flow rate and h represents the heating value per unit mass of fuel. This yields:

$$W_a = \frac{\overline{p'^2}}{\rho_0 c_0} 4\pi r^2 = \frac{(\gamma-1)^2}{4\pi\rho_0 c_0^5} f_c^2 \eta^2 (\dot{m}_F h)^2 \frac{V_{cor}}{V} \tag{32}$$

This expression has the same form as the power output of a distribution of uncorrelated monoples (Smith and Kilham (1963)) and requires a suitable estimate of the characteristic frequency f_c. Among the many possibilities explored in the literature, that used by Belliard (1997) and Rajaram et al. (2006) provides a suitable correlation of experimental data. The characteristic frequency is given by $f_c = U/L_f$ where L_f is a flame length defined as the point where the radially integrated intensity of CH* emission reaches a maximum and U is the average flow velocity at the burner exhaust. With this expression for the central frequency it is shown by the same authors that expression (32) provides a reasonably accurate scaling law for the radiated power (see figure 1). This is also consistent with early findings that the peak frequency of the combustion noise spectrum of turbulent flames is a linear function of both the flow velocity and the burning velocity and is inversely proportional to the burner diameter (Smith and Kilham (1963)).

4.1. Power spectral densities

The power spectral density of radiated sound may be determined by taking the Fourier transform of the correlation function:

$$R_{aa}(\tau) = \frac{(\gamma-1)^2}{4\pi\rho_0 c_0^5} \overline{\int_V \frac{\partial \dot{Q}'}{\partial t}(\boldsymbol{r}'_0 t) \frac{\partial \dot{Q}'}{\partial t}(\boldsymbol{r}'_0 + \boldsymbol{\xi}_0, t+\tau) d\boldsymbol{r}'_0 d\boldsymbol{\xi}_0} \tag{33}$$

One finds:

$$\begin{aligned} S(\omega) &= \frac{1}{2\pi} \int R_{aa}(\tau) \mathrm{e}^{i\omega\tau} d\tau \\ &= \frac{(\gamma-1)^2}{8\pi^2 \rho_0 c_0^5} \int (-\omega^2) R_{\dot{Q}'\dot{Q}'}(\boldsymbol{r}'_0; \boldsymbol{\xi}_0, \tau) \mathrm{e}^{i\omega\tau} d\tau d\boldsymbol{r}'_0 d\boldsymbol{\xi}_0 \end{aligned} \tag{34}$$

In this expression the space time correlation function of the nonsteady heat release fluctuations $R_{\dot{Q}'\dot{Q}'}(\boldsymbol{r}'_0; \xi_0, \tau)$ is a function of the space and time separations (ξ_0, τ) and in general of the position $\boldsymbol{r}'_0$. This function describes the spatial and temporal characteristics of heat release rate fluctuations in the reacting flow. It is difficult to give a general expression for this quantity but it is possible to continue the calculation in

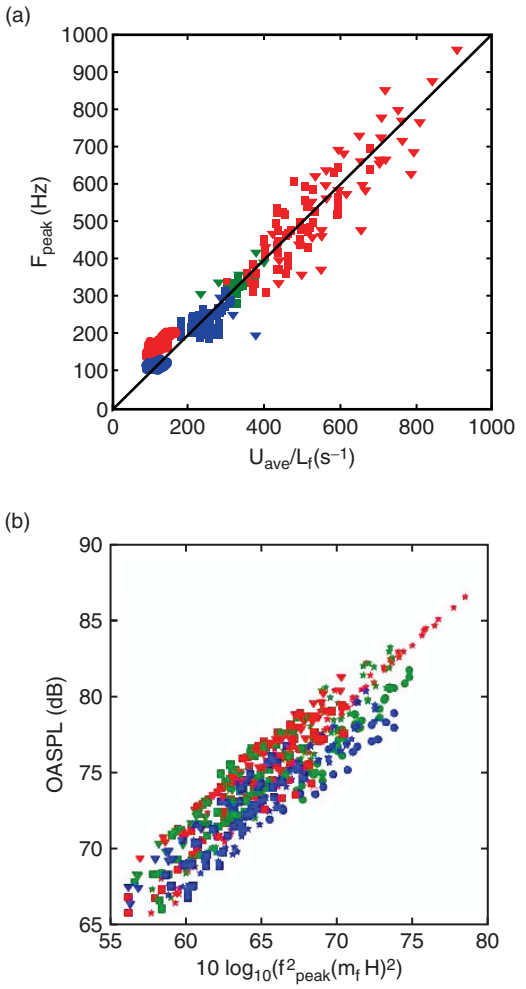


Figure 1: Premixed combustion noise correlations established by Rajaram et al. (2006). Red, blue, green symbols correspond to acetylene, natural gas, propane fuels respectively. Symbols correspond to different burner diameters (Circle) $D = 34.8$ mm, (Star) $D = 17.3$ mm, (Square) $D = 10.9$ mm, (Triangle) $D = 6.4$ mm. (a) Central frequency $f_c = f_{peak}$ correlation. (b) Radiated Over All Sound Pressure Level.

some special cases. For example, by restricting their analysis to premixed turbulent flames in the wrinkled flame regime, Clavin and Siggia (1991) were able to show that the rate of change of flame surface area followed a power law with respect to frequency and that the power spectral density of sound radiated by the flame featured a $f^{-5/2}$ dependence with respect to frequency. Experiments indicate that power spectral

densities of sound exhibit a range where the spectrum decays according to a power law but the exponent value changes from case to case. In the low frequency range one may assume that the power spectral density of the nonsteady heat release rate only weakly depends on frequency. The sound spectrum then follows a f^2 law. This seems to be well confirmed by experiments of Rajaram et al. (2006). The modeling of heat release rate fluctuations constitutes a central issue in the combustion noise analysis. In general these fluctuations are not easy to measure experimentally. In the premixed case one may link $\dot{Q}'$ to the light emission intensity of radicals like CH*, C^*_2 or OH* but the measurement is usually carried out over a line of sight and the fluctuation is integrated over a path through the flame. The method is not directly applicable in the nonpremixed or partially premixed cases.

One possible alternative explored by Boineau (1997) consists in relating $\dot{Q}'$ to the temperature fluctuations in the flow. This is accomplished by making use of a linearized version of the energy equation (13):

$$\rho c_p \frac{\partial T'}{\partial t} \dot{Q}' \tag{35}$$

This expression is obtained by neglecting diffusion fluxes and by suppressing the rate of change of the pressure perturbation $\partial p'/\partial t$. This last approximation is not easy to justify but allows a direct estimate of heat release fluctuations in terms of temperature fluctuations. It is convenient to introduce the relative fluctuation of temperature $\theta' = T'/\overline{T}$ which together with (35) and (34) yields a new expression for the radiated sound power spectrum:

$$S(\omega) = \frac{1}{8\pi^2} \frac{\rho_0 \omega^4}{c_0} \int R_{\theta'\theta'}(\boldsymbol{r}'_0, \boldsymbol{\xi}_0, \tau) \mathrm{e}^{i\omega\tau} d\tau d\boldsymbol{r}'_0 d\boldsymbol{\xi}_0 \tag{36}$$

where $R_{\theta'\theta'}(\boldsymbol{r}'_0, \xi_0, \tau)$ designates the relative temperature fluctuation space time correlation function. This function seems more accessible experimentally but it requires multipoint measurements of temperature fluctuations.

5. DIRECT COMBUSTION NOISE SOURCES

Mechanisms giving rise to sound emission from flames are analysed in this section by making use of simple experiments on modulated laminar flames. It is shown that intense sound emission may originate from flame interactions. Different flame geometries, stabilisation configurations and flow interactions were explored for premixed methane air flames in a mixture kept at constant equivalence ratio. Detailed analysis of the flame response to incoming flow modulations were carried by simultaneous measurements of velocity perturbations at the burner outlet, variations of the flame light emission and noise radiation. These signals are correlated using classical results of combustion noise theory. Phase conditional average images of the flame patterns are also acquired to examine the history of noise emission and locate the flame patterns corresponding to the instant of maximum noise production.

While the analysis is conducted for premixed laminar flames, implication for turbulent combustion noise analysis is also given. In a laminar configuration, the total heat release rate can be modelled as shown previously using the flame surface area $A(t)$ and the laminar burning velocity S_L: $\int_V \dot{Q} dV = \rho_f S_L(-\Delta h_f^0)A(t)$ where $-\Delta h_f^0$ is the enthalpy released by conversion of a unit mass of fuel. Considering now the turbulent flames, one may use flame surface density concepts and write the volumetric heat release rate in the form: $\dot{Q} = \rho f S_T\,(-\Delta h_f^0)\Sigma$, where Σ is the flame surface area per unit volume and S_T is the turbulent rate of conversion per unit flame area, which depends on the local conditions of flow and mixture properties. A budget for the flame surface density leads to a transport equation which takes the general form (Candel and Poinsot (1990)):

$$\frac{d\Sigma}{dt} = \epsilon\,\Sigma - \beta\,\Sigma^2 \tag{37}$$

The flame surface density increases when the flame is stretched ($\epsilon > 0$) by the turbulent flow, but this augmentation is limited by various processes (β is positive). Considering a unit volume of reactive mixture, equations (9) and (37) suggest that physical processes limiting the flame surface area locally produce sources of sound which are more intense than those associated with increasing flame surface area. The rate of change of the local flame surface area (and thus $\dot{Q}$) changes like the square of the flame surface area when this surface is being destroyed, while this rate of change of the flame surface area only increases linearly when flame area is being produced. The conclusion of this analysis is that physical processes leading to flame surface reduction should be mainly responsible for combustion noise production.

Among all mechanisms limiting the flame surface area, three types are studied in detail in what follows using controlled experiments in simple flow configurations. The first concerns extinction by thermal losses of a flame impinging on a cold boundary. The second involves mutual annihilation of neighboring flame elements, a mechanism which is suspected to limit the amount of flame surface area in turbulent combustors. The third case involves partial extinction of the flame due to large scale vortex structures convected in the outer shear layer formed between a premixed jet and the ambient atmosphere. As turbulent flames may be viewed as a collection of distributed monopoles, the dynamics and the sound produced by a collection of small conical flames anchored on a perforated plate are also compared to the dynamics and the sound output of a single conical flame.

The main objective is to prove that the noise radiated by these types of flame interactions, although diverse in nature, can mainly be attributed to the same mechanism of flame surface area destruction. The experimental configurations and diagnostics used to investigate these interactions are described in the next section. In the cases explored, the flames are modulated by an external perturbation to allow phase average conditional analysis of the flame motion and the radiated sound field. Results are presented for all geometries. Dynamics of the different flames during a cycle of modulation are first analyzed together with corresponding sound spectra of the radiated noise. The connection between the sound production and heat release fluctuations is then discussed. The analysis focuses on common features and differences between these cases.

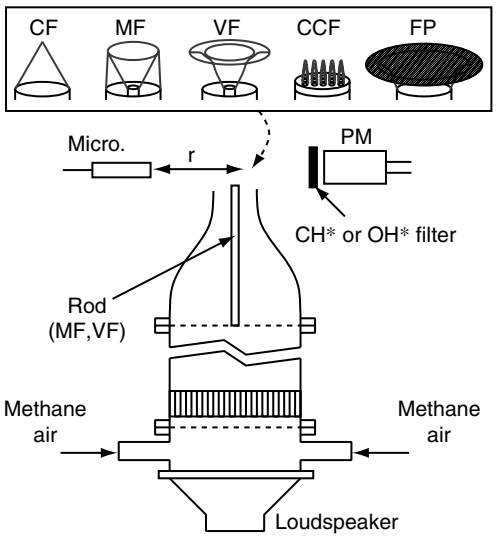


Figure 2: Experimental setup. PM: photomultiplier equipped with a CH^* or OH^* narrow-band filter. A loudspeaker is used to modulate the flow at the nozzle exit. Sound is recorded by a microphone (Micro.) at $r_\infty = 24–30$ cm away from the burner axis. The burner can be equipped with a central rod, a perforated plate, or a water cooled plate above de burner to modify flame shapes and stabilization zones (see Fig. 4). The burner is operated with a methane-air mixture $\Phi = 0.8 - 1.3$ in laminar regimes $\overline{\upsilon} = 1.2 - 5.0$ m s^{-1}.

5.1. Experimental setup

The experimental setup displayed in figure 2 comprises a burner with a d = 22 mm nozzle exit diameter and a loud-speaker fixed at its base. The body of the burner is a cylindrical tube of 65 mm inner diameter containing a set of grids and a honeycomb grid followed by a convergent nozzle of contraction ratio $\sigma = 16 : 1$. It is designed to produce a steady laminar flow with a flat velocity profile at the burner exit for mean flow velocities up to 5 m s^{-1}. Using this apparatus, velocity disturbances at the burner outlet are reduced to low levels as $\upsilon_{rms}/\overline{\upsilon} < 0.01$ on the central axis for free jet injection conditions. The burner is fed by the bottom with a mixture of methane and air. Experiments presented hereby were carried out at fixed equivalence ratios. As internal flow disturbances are minimised, the influence of external flow mo dulations can be investigated. These perturbations were produced by the loudspeaker driven by an amplifier, which is fed by a signal synthesizer. The flow was seeded with oil particles of a mean diameter around 2 μm and velocities at the burner nozzle exit were measured by Laser Doppler Velocimetry (LDV). Using this driving system, velocity modulations around the mean steady velocity like those presented in Fig. 3 were obtained at the burner exit plane for modulation frequencies up to 400 Hz. The time trace velocity presented in this figure is a purely harmonic signal as shown by the distribution of the Power Spectral Density (PSD) presented in the lower part of this figure with a frequency

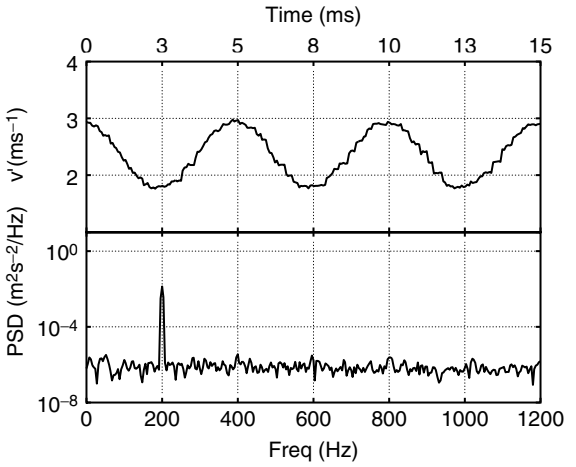


Figure 3: Velocity time trace during harmonic modulation of the flow at the burner outlet and the corresponding PSD distribution. "M" flame (Fig. 4, MF), $\Phi = 1.11$, $\overline{\upsilon} = 2.3$ m s^{-1}, $\upsilon_{rms} = 0.4$ m s^{-1}, $f_e = 200$ Hz.

peak at $f = 200$ Hz corresponding to the loudspeaker forcing frequency well above the background noise. For all flow conditions and driving frequencies explored, a minimum data rate collection of 15 kHz with 70% validation was ensured. Other records exhibit similar features to those presented in Fig. 3 with a strong peak at the driving frequency and a very low harmonic level. Incoming flow perturbations in the experiments presented below can thus be considered as purely sinusoidal at the nozzle exit for all the operating conditions explored.

For the range of flow velocities, equivalence ratios and burner geometries considered in what follows, laminar flames can be stabilized by different ways and take a wide variety of shapes. Five flame geometries are studied. Flames can for example be anchored on the burner lips and take a conical shape (Fig. 4, CF Statio), but the burner can also be equipped with a 2 mm diameter cylindrical rod placed inside the convergent unit and centered on the axis to modify flame freedom degrees. This rod provides an additional position for flame anchoring. The flame can be stabilized simultaneously on the central rod and on the external burner lips, and take an "M" shape (Fig. 4, MF Statio), but it can also be attached only to the central rod and adopt a "V" shape (Fig. 4, VF Statio). Stabilization as an "M" flame or a "V" flame can be obtained for the same flow velocity and equivalence ratio depending on the ignition location point. Once a regime is set, it remains stable without external intervention. The central rod can also be replaced by a perforated plate flush mounted at the top of the burner nozzle. This perforated plate features 2 mm in diameter regularly spaced holes to study collective effects of 25 small conical flames (Fig. 4, CCF Statio). A 10 mm thick water-cooled disc made of copper can also be placed above the burner nozzle. The plate surface is perpendicular to the axis and has a diameter $D = 100$ mm. A thermocouple set on the

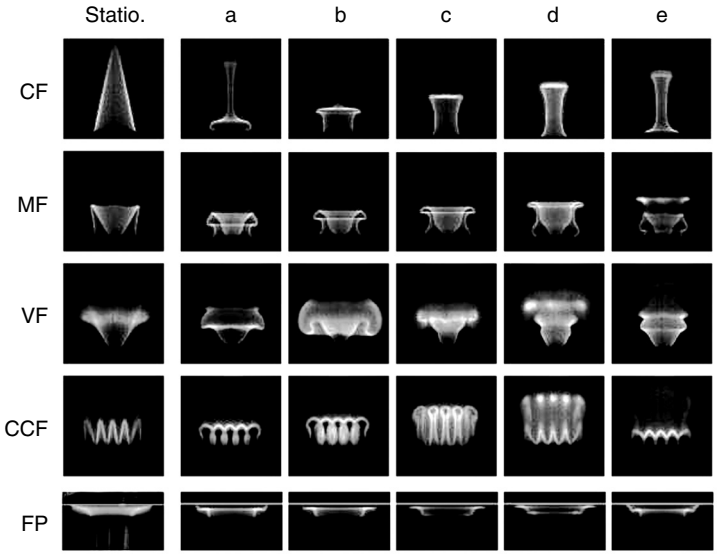


Figure 4: Flames snapshots. Statio.: stationary shapes. a-e: cycle of excitation. CF: conical flame. MF: "M" shaped flame. VF: "V" shaped flame. "CCF": collection of (small) conical flames. FP: Flame plate interaction.

axis of the disc is used to control the plate temperature, which was roughly kept constant in all experiments at a value above the wet bulb temperature. The plate can be moved vertically above the burner and the flame may adopt different shapes (Schuller et al. (2002); Fernandes and Leandro (2006)). For the configuration studied hereby, the plate height above the burner is fixed at $z = 7.6$ mm and the flame is stabilized on the outer periphery of the burner with fresh reactants impinging the plate in the center region. The flame stabilized on the burner lips extinguishes at its other extremity by thermal losses of peripheral flame elements in regard to the cooled plate. Experiments were carried out on laminar flames maintained at a fixed equivalence ratio in forced flow configurations.

It is important to mention that all of these flames also develop self-sustained oscillations without the need of external forcing, but these regimes are not studied here. During forcing of the flow, the flame oscillates and responds by noise emission. The sound radiated is measured with the microphone *Micro*. placed at a distance $r_\infty = 24$ to 30 cm away from the burner axis depending on the case studied. This distance is not very large compared to the typical dimension of the reactive region and one cannot consider that compactness and far field conditions are completely fulfilled. This choice was made to minimize reflections on solid boundaries and on the neighboring equipments. As will be shown later, the observation distance is sufficient for the present purpose. Heat release fluctuations are also measured using a photomultiplier PM equipped with a CH^* or a OH^* narrow band filter. The current originating from the PM is proportional to the light emitted I by free radicals present in the flame reaction zone. For homogeneous mixtures,

fluctuations of this current are directly proportional to variations of the heat release rate. This relationship ceases to be valid when the mixture is inhomogeneous. LDV, PM and Micro output voltages were simultaneously recorded and acquired on an A/D board piloted by a computer at an acquisition frequency f_a = 16 384 Hz during a 2-second recording time. This duration enables to acquire at least 100 periods for the lowest driving frequency f_e = 50 Hz and 800 periods for the highest frequency f_e = 400 Hz. These informations are processed to extract delayed time rates of change of the flame light chemiluminescence using eq. (8). As flow interactions are laminar, records were reproducible from cycle to cycle. Power spectral densities were calculated using Welch's periodogram method. Power spectral densities (PSD) and sound pressure levels (SPL) were computed using the following definitions: $PSD(dB) = 10 \log_{10} [PSD(Pa^2/Hz)\Delta f/p_{ref}^2]$, where Δf is the spectral resolution, $p_{ref} = 2.10^{-5}$ Pa is a reference acoustic pressure and $SPL(dB) = 20 \log_{10} (p_{rms}/p_{ref})$.

Velocity, flame emission and acoustic signals were recorded together with phase averaged images of the flame patterns at regularly spaced instants during the driving cycle using an intensified CCD camera. Each image is formed by averaging 100 snapshots with an exposure time Δt = 100 μs (Fig. 4 CF, MF, VF, CCF, FP a-e). As configurations explored are laminar and the exposure time small compared to the driving period $T_e = 1/f_e$ (T_e = 2.5 to 20 ms), each phase averaged image can be considered as a snapshot of the flame pattern at a precise instant in the driving cycle. Cycle to cycle reproducibility can also be assessed by examining the image sharpness. Except some blurring effects on the flame periphery of the stationary "V" shape flame (Fig. 4, VFa-e) and neighbouring inner reaction zones of the collection of small conical flames (Fig. 4, CCFa-e), other phase averaged images are perfectly sharp and one can easily follow the flame contour. For each driving frequency, 21 images at regularly spaced phases were taken to cover a complete excitation cycle and post-processed to extract the flame contour. The camera line of sight crosses a row of holes to get a clear image for the collection of conical flames. Except for this configuration, flame surface areas were computed assuming a cylindrical symmetry. Using eqs. (4) and (9), relative flame surface area fluctuations over a cycle can be directly compared to relative flame light variations and noise radiation. Operating conditions are summarized in Table 1.

It was not possible to obtain exactly identical flow and modulation operating conditions for all the configurations explored. This modulation level was adjusted to get reasonable signal to noise ratios for the measured signals. However, orders of magnitude were kept constant.

5.2. Spectral and time traces analysis

Without forcing, flames are stable for the operating conditions explored. Except for the V-flame (Fig. 4, VF), there is no visible motion of the front. In the VF case, a small disorganized motion of the flame limited to its free periphery is observed. This motion can be attributed to interactions with small scale vortices convected in the mixing layer between the premixed jet and the surrounding air. The mean background noise level in the laboratory with all apparatus on is lower than 60 dB. Without combustion, the sound level does not exceed 65 dB when the flow is submitted to modulations. This is also the

Table 1: Summary of operating conditions. Φ: equivalence ratio, f_e: modulation frequency, *SPL*: Sound Pressure Level, $\bar{A}$: mean flame surface area, A_{rms}: flame surface area fluctuation, $\bar{v}$: mean velocity at the burner outlet, and v_{rms}: imposed velocity fluctuation level.

	Φ	f_e (Hz)	SPL (dB)	$\bar{A}$ (m^2)	A_{rms} (m^2)	$\bar{v}$ (m s^{-1})	v_{rms} (m s^{-1})
CF	1.11	50	73	1.8 10^{-3}	1.9 10^{-4}	1.7	0.8
MF	1.13	150	91	–	–	1.71	0.50
	1.11	200	92	2.1 10^{-3}	4.0 10^{-4}	2.3	0.4
VF	0.8	150	–	–	–	1.87	0.15
	0.92	170	83	–	–	2.56	0.30
	0.92	170	–	–	–	2.56	0.12
	1.11	100	96	2.1 10^{-3}	9.0 10^{-4}	2.3	0.6
CCF	0.95	400	96	–	–	5.0	0.9
FP	0.95	200	84	–	–	1.20	0.36

maximal value of the sound level measured by the microphone with combustion but in the absence of modulation, except for the V-flame case (Fig. 4, VF). The sound level is slightly louder and can be attributed to the flame motion at its periphery. But sound levels measured in the absence of external forcing of the flow are relatively low and the flames can be considered as quiet. The situation is quite different in the presence of external modulation. The flames respond with a periodic motion accompanied by a strong noise emission. A qualitative analysis combining a description of the flame patterns at successive phases over a cycle of excitation and the corresponding sound spectra is now carried out to understand the dominant physical mechanism leading to noise emission.

Unsteady flame-plate interaction is first investigated. The motion is periodic with a frequency of oscillation equal to the forcing frequency f = 200 Hz. The velocity perturbation at the burner outlet produces a deformation of the flame front (figure 5a) at the burner lips, which is then convected by the flow towards the plate (figure 5b-c). Two wrinkles following each other are visible for this case. As these perturbations approach the plate, the flame surface extends along the plate and it increases smoothly up to a maximum value corresponding to the image above the last one in figure 5c. In the next image (figure 5d), a large portion of the flame front has disappeared and the flame surface area is reduced compared to that available in the previous one. This suggests that a sudden drop of flame surface area has occurred and that a large negative rate of change of heat release has taken place. Flame extinction is due to thermal losses at the cold boundary which is water cooled and has a temperature maintained at T ~ 300 K. The flame in the snapshot figure 5d has nearly retrieved the initial shape shown in the top image of figure 5a. Because images were taken at regularly spaced instants, this suggests that the rate of change of heat release takes higher value during phases of flame destruction in the cycle than during the smooth increase of the flame surface area (Schuller et al. (2002)).

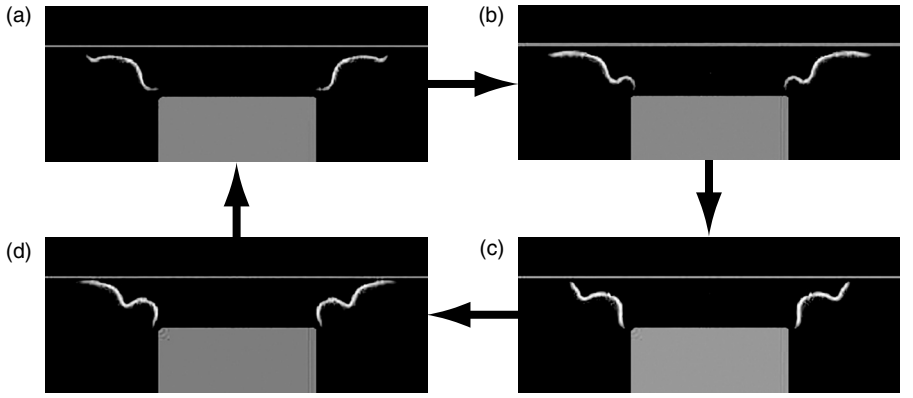


Figure 5: Cyclic flame motions during flame interaction with a cold boundary (Fig. 4, FP), $\Phi = 0.95$, $\overline{\upsilon} = 1.20$ m s^{-1}, $\upsilon_{\text{rms}} = 0.36$ m s^{-1}, $f = 200$ Hz.

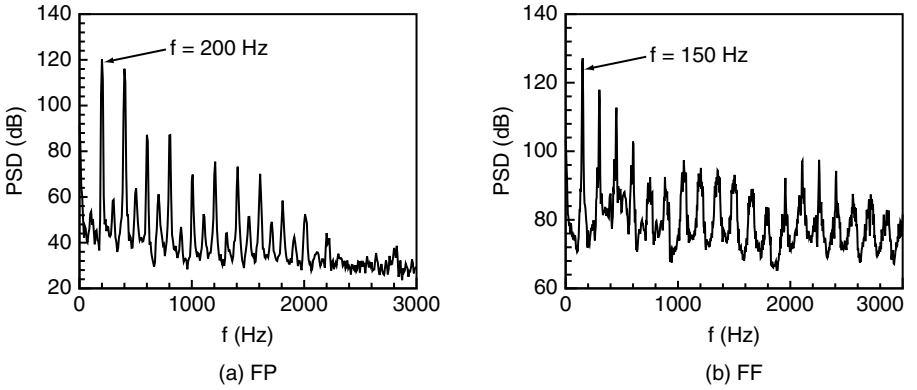


Figure 6: Sound emission power spectral density during (a) Flame-Plate interaction Fig. 5, $\Phi = 0.95$, $\overline{\upsilon} = 1.20$ m s^{-1}, $\upsilon_{rms} = 0.36$ m s^{-1}, $f = 200$ Hz and (b) Flame-Flame interaction Fig. 7, $\Phi = 1.13$, $\overline{\upsilon} = 1.71$ m s^{-1}, $\upsilon_{\text{rms}} = 0.50$ m s^{-1}, $f = 150$ Hz, $r_{\infty} = 0.25$ cm.

The overall noise radiated recorded in the laboratory reaches in the present case 84 dB, i.e. about 20 dB more than without upstream perturbation, which means that the rms pressure fluctuation in the laboratory is more than 10 times greater with modulation than without. Repeating this experiment, without the plate or without combustion, the noise level falls to a value close to sound level without modulation ($\sim$ 60 – 65 dB), implying that the noise source is directly related to the flame-plate interaction (Schuller et al. (2002)). The noise spectrum corresponding to this flame-plate interaction is plotted in figure 6a. Several well defined peaks are present. The first peak corresponds to the forcing frequency $f = 200$ Hz which is also the frequency of the flame oscillation. The other

peaks correspond to harmonics of this fundamental frequency. The harmonic energy content is non negligible and for some cases can exceed that concentrated at the forcing frequency. In some situations not described here, peaks at half the fundamental frequency are also visible (Fernandes and Leandro (2006)). Pressure fluctuations have the same period as velocity fluctuations at the burner outlet, the latter are purely harmonic whereas pressure fluctuations have a richer harmonic content. This suggests a strong non linear process in the transfer of energy between outlet velocity fluctuations and the sound radiation.

Thermal losses during unsteady flame-wall interactions may constitute an intense source of combustion noise, for example during extinctions of large coherent structures entrained by the flow and impacting on solid boundaries (Smith and Zukoski (1985); Kendrick et al. (1996)), or if the turbulent flame is simply stabilized close to a wall. However in most combustors the flame is stabilized away from the boundaries and may not reach the chamber walls. Other sources of combustion noise should be explored as for example mutual flame interactions.

These interactions are investigated for a flame stabilized over the burner lips and the central rod (MF in figure 4). The "M" shape adopted by the flame is well suited to studies of flame-flame interactions which may take place between neighboring branches (Schuller et al. (2003)). The case presented here corresponds to an equivalence ratio $\Phi = 1.13$, a mixture flow velocity $\overline{\upsilon} = 1.13\text{m s}^{-1}$, a modulation level fixed to $\upsilon_{rum} = 0.50\text{ m s}^{-1}$ and a modulation frequency $f = 150$ Hz. The description of the flame motion over a cycle of excitation starts as in the flame-plate interaction. A velocity perturbation originates at the burner lips and produces a deformation of the flame front at the base of the burner (figure 7a). The perturbation mostly affects the outer branch of the "M" flame. It is then convected by the mean flow towards the top of the flame. As the deformation travels along the flame front, the two branches of the "M" are stretched in the vertical direction and get closer (figure 7b-c), up to an instant in the cycle where the flame surface area is maximum and two flame elements interact (figure 7d). The outcome of this mutual annihilation depends on the spatial position of the first interaction. In some cases pockets of fresh reactants may be trapped in a flame torus while in other cases this will not occur. For some operating conditions not shown here, up to two flame toruses can be produced. During interaction of these flame elements the shape of the reactive front undergoes a strong alteration. As in the flame-plate situation, after the mutual interaction, the flame quickly retrieves its initial shape at the beginning of the cycle (figure 7a). The short phase in the cycle during flame surface destruction produces a faster rate of change of the flame surface area than the longer phase of flame surface production by stretch. The same mechanism operates as in the flame-plate interaction, except that flame surface destruction is produced by mutual annihilation of neighboring front elements and not by thermal losses.

The overall sound pressure level reaches 91 dB. The pressure spectrum in figure 6b is quite similar to that associated with the flame-plate interaction in figure 6a. The presence of harmonics of the fundamental frequency $f = 150$ Hz indicates that the pressure signal is also periodic with an oscillation frequency corresponding to the flame oscillation frequency which is also the forcing frequency. However the wave shape is nonlinear

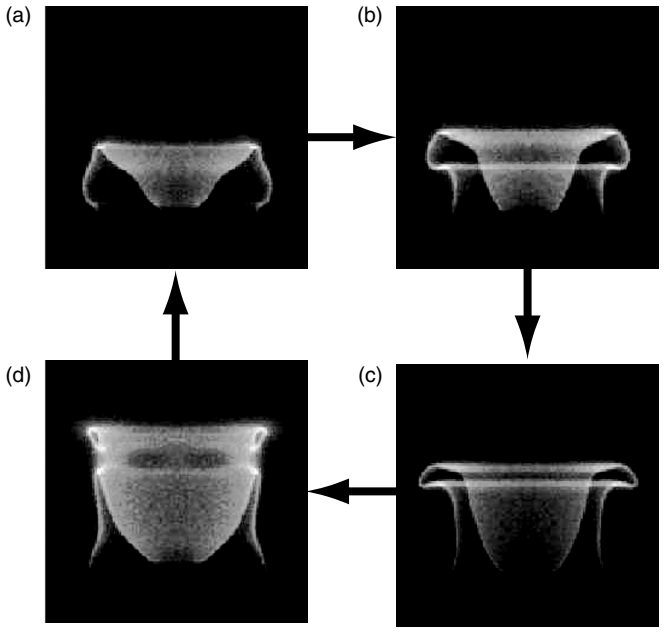


Figure 7: Cyclic flame motion during mutual flame annihilation (Fig. 4, MF), $\Phi = 1.13$, $\overline{v} = 1.71$ m s^{-1}, $v_{rms} = 0.50$ m s^{-1}, $f = 150$ Hz.

with a rich harmonic content. These energetic harmonics indicate that the physical process at the origin of the noise involves a rapid change of the rate of heat release.

As mutual flame annihilation is believed to control and to limit flame surface area in turbulent combustion (eq. (37)), the previous findings suggest that this mechanism could also be the main source of noise in turbulent combustors.

The third mechanism discussed in this article involves flame-vortex interactions. This is envisaged in an inverted conical flame configuration (VF in figure 4). The flame is now stabilized on the central rod and extends freely from this central body. Organized toroidal vortical structures are created in the outer mixing layer of the jet exhausted by the burner. In this layer, atmospheric air is entrained by the inner jet of premixed reactants. Operating conditions for the case under consideration are $\Phi = 0.8$, $\overline{v} = 1.87$ m s^{-1}, $v_{rms} = 0.15$ m s^{-1} and a forcing frequency $f = 150$ Hz. The mechanism of vortex shedding and its interaction with the flame front is briefly discussed. When a velocity perturbation is produced at the burner outlet a vortical structure is shed from the outer rim and is convected towards the flame front at a velocity approximately equal to half the mean flow velocity (Birbaud et al. (2007)). Images in figure 8 display both the vortical field obtained by PIV measurements and the corresponding flame position in the plane of symmetry of the burner (Durox et al. (2005)). The vortex shedding

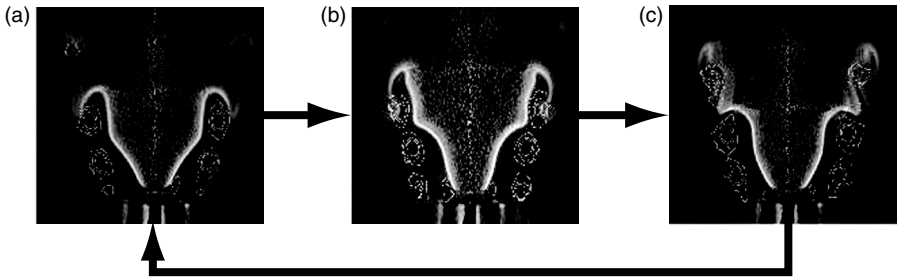


Figure 8: "V" - flame motion during cyclic modulation of the flow (Fig. 4, VF), $\Phi = 0.8$, $\overline{v} = 1.87$ m s^{-1}, $v_{rms} = 0.15$ m s^{-1}, $f = 150$ Hz.

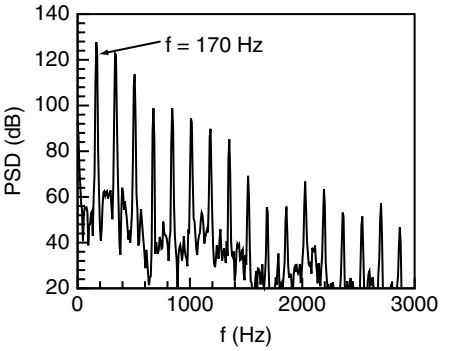


Figure 9: Sound emission power spectral density during flame-vortex interaction. $\Phi = 0.92$, $\overline{v} = 2.56$ m s^{-1}, $v_{rms} = 0.30$ m s^{-1}, $f = 170$ Hz, $r_\infty = 24$ cm.

frequency is synchronized with the acoustic forcing frequency $f = 150$ Hz. Travelling away from the burner lips, these vortices interact with the flame by rolling up the reactive front (figure 8a). Maximum roll-up is obtained close to the free periphery of the flame when the vortex reaches the flame sheet (figure 8b). At some points, flame stretch induced by the vortex is so strong that the flames extinguishes (figure 8c). A large portion of flame suddenly vanishes during this interaction in a short period of time.

The spectra examined in figure 9 is not associated with images displayed in figure 8, but the flame modulated at a slightly higher frequency $f = 170$ Hz features the same patterns due to vortex enrollment and extinction. The overall sound radiated is 83 dB. As for the two previous cases, the power spectral density also features a broad harmonic content with a fundamental frequency corresponding to the flame motion oscillation frequency $f = 170$ Hz, indicating a strongly non linear mechanism at the origin of the noise production.

5.3. Discussion

The qualitative analysis developed in the previous section is extended to infer correlations between the sound radiation and the combustion dynamics. This is be achieved by comparing time evolutions of the pressure fluctuation p' and the light emission I signals recorded during the three experiments. Results are respectively displayed in figures 10 to 11 for the flame-plate (FP configuration), flame-flame (MF configuration) and flame-vortex (VF configuration) interactions. The relative intensity $I/\overline{I}$ of the CH^* natural flame emission signal recorded by the PM is plotted at the top of these figures. The corresponding pressure fluctuations p' recorded by the microphone *Micro*. are also plotted at the bottom of these figures with the delayed time rate of change dCH^*/dt. This latter signal is obtained by post-processing the CH^* signal. The first step is a differentiation and a low pass-filtering. It takes into account the time delay $\tau \simeq r/c_0$ for the sound wave to cover the distance r between the compact flame and the microphone M, with c_0 taken equal to 343 m s^{-1}. The resulting signal is then scaled to get a signal $dCH^*/dt(t)$ with the same amplitude as the pressure fluctuation $p'(t+\tau)$ by setting $\rho_\infty/(4\pi r)(\rho_u/\rho_b-1)k = 1$ in eq. (4). Owing to the combustion noise theory, these quantities $dCH^*/dt(t)$ and $p'(t+\tau)$ plotted in figures 10 to 11 should be equal for a homogeneous mixture maintained at a constant equivalence ratio. Except some differences in the high frequency content, the signals obtained for high levels of velocity modulation are well correlated for the three cases explored (figures 10a, 10b and 11a). It is in particular shown that the phase corresponding to the largest negative pressure fluctuation always corresponds to the maximum negative rate of change of the CH^* signal.

Concerning the MF, phases corresponding the conditioned images of figure 7 are also plotted in figure 10b as circles. These data were obtained by directly measuring the relative flame surface area fluctuations $A(t)/\overline{A}$ using the snapshots of figure 7 and by assuming an axisymmetric configuration. During a cycle of oscillation, the flame light emission level increases slowly when the flame surface increases up to its maximum value. It then drops suddenly to its minimum value on a shorter time scale than during the flame surface augmentation period. The maximum rate of decrease of the CH^* signal fixes the pressure fluctuation amplitude level in the three cases explored, showing that flame surface destruction is the main mechanism leading to combustion noise production. When coupled with a resonant acoustic mode of the burner, theses pressure fluctuations are strong and robust enough to generate self sustained combustion oscillations (Durox et al. (2002); Schuller et al. (2003); Durox et al. (2005)). The same relation eq. (4) between far field pressure fluctuations and heat release variations holds in these unforced configurations.

The last case involving a flame-vortex interaction with an "V"-flame (VF) slightly differs from the two previous cases (FP and MF). Results not presented here show that during self-sustained situations the level of velocity fluctuation at the burner outlet takes lower values than the value used in the forced flow configuration. The time rate of change of the light emission signal dCH^*/dt and the corresponding pressure fluctuation p' are not well correlated anymore for self-sustained oscillations (Durox et al. (2005)) or in forced configurations for a low modulation level as in figure 11b. It is believed that this phenomenon is due to the mixing layer between the jet and the surrounding air

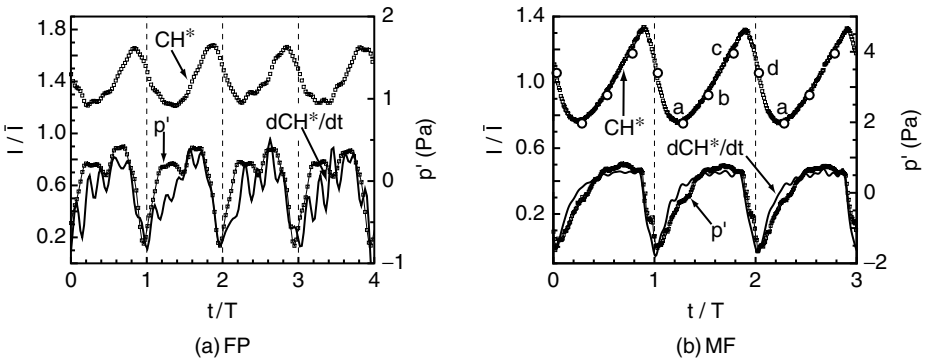


Figure 10: Time history of acoustic pressure and heat-release during (a): flame-plate interaction (FP), $\Phi = 0.95$, $\bar{\upsilon} = 1.20$ m s^{-1}, $\upsilon_{\mathrm{rms}} = 0.36$ m s^{-1}, $f = 200$ Hz and (b): flame-flame interaction (MF), $\Phi = 1.13$, $\bar{\upsilon} = 1.71$ m s^{-1}, $\upsilon_{rms} = 0.50$ m s^{-1}, $f = 150$ Hz, $r_\infty = 0.25$ cm.

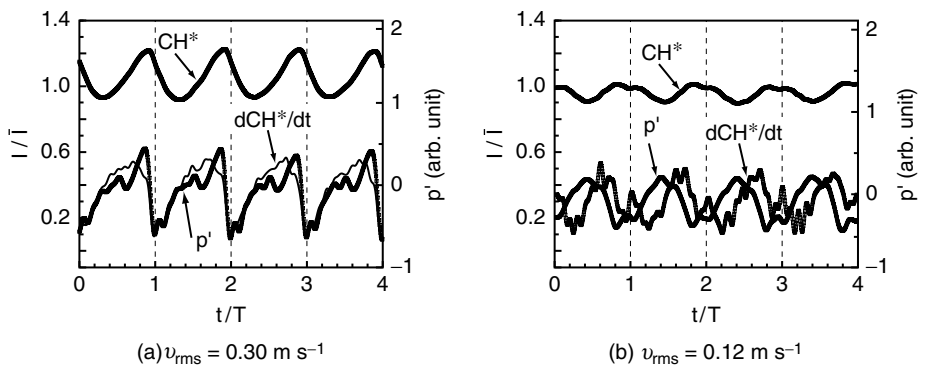


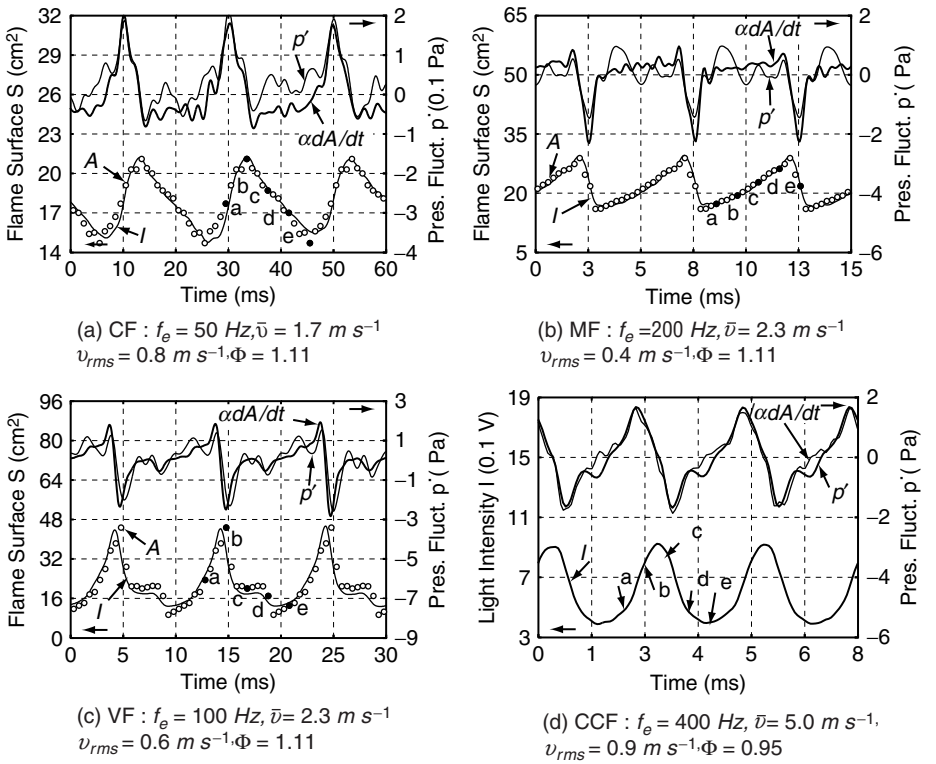
Figure 11: Time history of acoustic pressure and heat-release during the flame-vortex interaction (ICF), $\Phi = 0.92$, $\bar{\upsilon} = 2.56$ ms^{-1}, $f = 170$ Hz, $r_\infty = 24$ cm. (a): Moderate modulation level $\upsilon_{rms} = 0.30$ m s^{-1}. (b): Low modulation level $\upsilon_{\mathrm{rms}} = 0.12$ ms^{-1}.

where the mixture composition varies. For premixed systems, heat release rate fluctuations can be due to changes of the flame surface area (A) and to changes in the rate of conversion per unit flame area (S_L). Neglecting all other effects, the rate of consumption of reactants mainly depends on the local equivalence ratio Φ in laminar cases. For the FP and FF interactions, the local mixture composition was kept constant. Fluctuations of the heat release rate were only due to fluctuations of the flame surface area, a situation where eq. (9) is valid. In the VF case, heat release fluctuations can also be due to variations in the local consumption rate and eq. (9) may no longer be suitable.

Indeed, this equation is not satisfied for small velocity modulations (figure 11b and Durox et al. (2005)), but the correlation improves when the modulation level is raised as shown in figure 11a. The reason could be that increasing the velocity modulation level promotes flame surface fluctuations (sensitive to velocity fluctuations) compare to fluctuations of the mixture composition (insensitive to velocity fluctuations). For large velocity modulation levels mixture composition effects may be neglected in the mechanism of noise production (figure 11a), whereas it is not the case for lower velocity fluctuations (figure 11b and Durox et al. (2005)).

Other cases are presented in the next figure where attempts were made to identify the instant of strong noise production. This was done by comparing the flame surface area extracted from the phase averaged images at regularly spaced instants in the oscillation cycle and the PM signal. For the sake of simplicity, light emission signals were superposed with the same scale on the flame surface area plots. This enables to identify the instant of maximum noise production in each images sequence.

The first flow interaction is that of a conical flame submitted to convective waves (Fig. 4-CF, a-e). Under strong harmonic forcing $v_{rms}/\overline{v} = 0.47$, the shape of the flame exhibits periodic wrinkles with a large bulge convected from the burner lips to the flame tip inducing moderate flame surface variation $A_{rms}/\overline{A} = 0.11$. Nevertheless, the creation of this bulge leads to flame surface production at the flame base (Fig. 4-CF, a). This instant corresponds to a positive pressure peak in the time trace in Fig. 12-CF, but the sound level 73 dB in Tab. 1) remains moderate compared to the laboratory background noise 60 dB. Noise is mainly associated with phases of flame surface production. The sequence of images in Fig. 4-MF corresponds to an "M" shaped flame submitted to the same harmonic forcing, but it exhibits a different response than the conical flame. As the wrinkle travels towards the top of the flame, elements of neighboring fronts get closer to each other and pinch off (Fig. 4-MF, instants c and d). This mutual annihilation leads to the creation of a flame torus, which is convected downstream. The net result of this cyclic interaction is a flame surface variation $A_{rms}/\overline{A} = 0.19$ of the same order as the incoming flow perturbation level $v_{rms}/\overline{v} = 0.17$. The instant of flame pinching (Fig. 4-MF, e) corresponds to a large negative pressure peak (Fig. 12-MF), with an overall SPL equal to 92 dB (Tab. 1). In contrast to the previous case, noise is mainly associated to the fast rate of annihilation of the flame surface area. This interaction results in a large sound level even for a relatively low flow modulation level. In the cycle presented in Fig. 4VF, the rolling up motion at the flame periphery is due to a toroidal vortex shed from the burner lips generated by the flow perturbation at the burner outlet $v_{rms}/\overline{v} = 0.26$. This vortical structure is convected towards the flame front at half the mean flow velocity in the mixing layer formed at the interface between the premixed jet and the surrounding air. As flame elements are rolled up at the front periphery, they are stretched, mutually interact (Fig. 4-VF, b-c) and contribute to a large rate of destruction of the flame surface area $A_{rms}/\overline{A} = 0.43$. This event corresponds to an intense negative pressure peak in Fig. 12-VF between instant b and c and results to a total SPL radiated of 96 dB. The last interaction explored involves an ensemble of small conical flames submitted to acoustic modulation that will be described in details below. Quantitative comparisons are now carried out.



Figuer 12: Time traces: OH* light intensity I, flame surface area A, pressure fluctuations p' and computed pressure fluctuations $\alpha dA/dt$. Circles indicate extracted flame surface areas A in cm^2. Black circles or arrows marked *a, b, c, d* correspond flames patterns presented in images from Fig. 4.

Flame surface area A extracted from phase averaged images and indicated as circles in cm^2 are presented in Fig. 12 together with flame light intensity I displayed as solid lines. Emission signals I were rescaled using $I'/\bar{I} = A'/\bar{A}$ which remains valid as long as fluctuations of the heat release rate are solely due to fluctuations of the flame surface area. Both signals perfectly coincide during all phases of the driving cycle indicating that the flame dynamics can be well retrieved by examining OH* radical light emission during flame interactions with convective waves (Fig. 12-CF), mutual annihilation of flame elements (Fig. 12-MF) or flame vortex interactions (Fig. 12-VF). Time traces reveal strong non linearities during short phases of the driving cycles leading to either large rates of surface production (CF) or surface destruction (MF, VF). The pressure p' radiated in the laboratory measured by the microphone can be compared to the theoretical pressure fluctuation computed from the delayed rate of change of the flame surface area dA/dt, or equivalently from dI/dt, using eqs. (4) and (9). Results are plotted at the top of Fig. 12 where the coefficient $\alpha = \rho_\infty/(4\pi r)(\rho_u/\rho_b - 1)S_L$ was computed

using values indicated in Tab. 1. Pressure peak positions, amplitudes and rms levels are correctly retrieved for the conical, "M" and "V" shaped flames submitted to convective waves (Figs. 12-CF, MF and VF). It should be noted that these results obtained for a microphone located in the nearfield of the flame still obey the classical relation derived for farfield conditions.

The last interaction involving a collection of small conical flames differs to some extent from previous cases because phases of flame surface creation and destruction take approximatively the same duration (Fig. 12-CCF). Instants of maximum rate of production and destruction correspond respectively to a positive and negative pressure peak of approximatively the same amplitude in pressure time trace in Fig. 12-CCF. This observation differs from some of our previous expectations (Candel et al. (2004)), that mechanisms leading to flame surface destruction were the main contributions to the overall noise production. Here, both creation and destruction phases result in opposite large rate of change of the flame surface area associated to intense pressure peaks. The flame surface area evolution was not extracted because flame contours are blurred in many images in Figs. 4-CCF a-e due to the combined motion of each individual flame. Nevertheless, the flame dynamics can be described using flame emission signals. Letters in Fig. 12-CCF indicate instants corresponding to images in Fig. 4-CCF. The mean flow velocity at the plate holes outlet is $v = 5.0$ m s^{-1} and the relative perturbation level $v_{rms}/\overline{v} = 0.16$ remains moderate. Flames first stretch and elongate quickly (Figs. 4-CCF a and b) leading to large flame surface area production. Flames then suddenly break up and release small spherical pockets of fresh reactions (Figs. 4-CCF c and d). It is interesting to note that for about the same flow perturbation level $v_{rms} = 0.8 - 0.9$ m s^{-1}, the SPL radiated by the collection of small conical flames reaches 96 dB, whereas the single conical flame remains relatively quiet (CF, SPL = 73 dB, Tab. 1). It should also be mentioned that this result was obtained for a reduced mass flow rate $\dot{m} = 0.29$ g s^{-1} for the CCF configuration to be compared with $\dot{m} = 0.59$ g s^{-1} consumed by the single conical flame (CF). The ensemble of small conical flames is a more compact configuration than the single conical flame leading to higher flame surface densities and to stronger noise production. It is a special case where sound production associated with rates of flame surface creation by stretching flame elements are of the same order than rates of flame surface destruction during mutual annihilation (eq. 37).

The previous examples indicate that rapid rates of change of flame surface area constitute a strong source of combustion noise, a mechanism which could be the source of much of the noise radiated by turbulent flames. Events corresponding to large and fast rates of variation of the flame surface area were also identified as the main contribution to the overall radiated noise.

Flame-wall interactions, mutual annihilations of flame elements and to some extent flame-vortex interactions correspond to mechanisms limiting the flame surface area. It was shown here that these processes may induce large negative rates of change of the heat release rate $d\dot{Q}/dt$. Values of this quantity are larger during phases of flame surface destruction than during phases of flame surface production, implying that flame surface dissipation is the dominant mechanism of combustion noise production in combustors operating in a continuous mode (gas turbine combustors, industrial boilers, furnaces...).

This is so because extinction or dissipation of flames are usually faster than flame surface generation. However, it is important to note that there exist also other mechanisms leading to noise production during flame stretch, collective effects and flame surface creation. In automotive engines for example, where combustion noise mainly originates from the violent ignition of the reactants introduced in the cylinder, the fast rate of production of flame surface area is the dominant mechanism.

These flame interactions were studied in configurations open to atmospheric condition. Sound is only radiated from the flame towards the farfield or eventually interacts with the upstream burner. Such interactions may also drive instabilities under resonant conditions with the upstream manifold as shown in the studies (Durox et al. (2002); Schuller et al. (2003); Durox et al. (2005); Noiray et al. (2006)). Considering that flames open to atmospheric conditions constitute relatively weak noise sources compared to enclosed flames (Mahan and Karchmer (1991)), one may except that these flame interactions also largely contribute to broad band noise production in a confined environment.

6. CONFINEMENT EFFECTS, NOISE AND INSTABILITIES

Most of the work on combustion noise has concerned radiation from unconfined flames. This case is less dependant on the geometry because sound generation is not modified by reflection from boundaries. It is also easier to examine unconfined flames with optical techniques. However, in many applications combustion takes place in confined environments and radiation takes place from the combustor inlet or exhaust sections. Another important issue in these confined configurations is the interactions of sound pressure waves with their sources. This generally involves two types of interactions:

- The first correspond to weak coupled interactions manifested by enhancement of sound radiation from confined systems by the presence of discrete peaks in the radiated sound spectrum. A typical core noise spectrum measured in the far field of an engine has the general form of an open flame spectrum, but with a superimposed series of relatively sharp peaks corresponding to the resonant frequencies of the engine (Mahan and Karchmer (1991)). Combustion under these conditions remains globally stable and interactions between waves and combustion noise sources are relatively weak. Pressures waves generated by the flame and reflected back towards the combustion region from the boundaries may be scattered by the flame (Lieuwen and Cho (2005)) and one expects that the turbulent combustion region to be only weakly modified by acoustic perturbations in these regimes.
- The second type pertains to strongly coupled interactions when acoustic energy densities fed into some of the burner eigenmodes are high enough to destabilize the reactive flow under some phase match conditions. The flow inside the chamber then generally evolves into a self-sustained oscillations accompanied by strong noise production inside the chamber (Candel (2002)) also leading to modifications of the radiated spectrum in the farfield region. In these strongly coupled regimes, acoustic fluctuations and large coherent structures of the reactive flow are synchronized around a resonant frequency of the burner. The difference between these regimes and the weakly coupled ones is that now pressures waves generated by the

flame carry enough acoustic energy densities to modify the flow and affect the combustion process. As flames are to the leading order insensitive to pressure perturbations (Dowling (1995); Truffin and Poinsot (2005)), acoustic-flame coupling mechanisms are indirect. Pressure waves generated from the flame are first converted into flow perturbations through interactions with the system boundaries (inlets, outlets and walls) leading to modifications of the fuel/mixture proportions or to unsteady velocity fields in the region upstream to the combustion region (Ducruix et al. (2003)). These flow perturbations are then transported towards the flame, modifying the heat release rate and thus affecting noise production in the flame region. In these regimes, unsteady flow, combustion and acoustics are intimately coupled.

A general framework to study confinement effects is to expand the pressure field on a basis of eigenmodes. This method has been extensively exploited in the area of combustion instabilities and it is equally applicable in the present context (Culick and Yang (1995)).

$$p'(\boldsymbol{r},t) = \sum a_n(t)\psi_n(\boldsymbol{r}) \tag{38}$$

where the normal modes satisfy the homogeneous Helmholtz equation:

$$\boldsymbol{\nabla}\cdot\bar{c}^2\boldsymbol{\nabla}\psi_n + \omega_n^2\psi_n = 0 \tag{39}$$

together with suitable conditions at the boundaries:

$$\boldsymbol{\nabla}p' \cdot n = f \tag{40}$$

In these equations ω_n designates the $n-th$ eigenvalue of this boundary value problem and n is the outward normal unit vector. The amplitudes in the modal expansion (38) are easily obtained from a set of ordinary differential equations:

$$\frac{d^2a_n}{dt^2} + \omega_n^2 a_n = F_n \tag{41}$$

where

$$F_n = \frac{\bar{c}^2}{\Lambda_n}\left[\int_V \frac{\gamma-1}{\bar{c}^2}\frac{\partial\dot{Q}'}{\partial t}\psi_n(\boldsymbol{r}_0)dV(\boldsymbol{r}_0) + \int_S f\psi_n(\boldsymbol{r}_0)dS(\boldsymbol{r}_0)\right] \tag{42}$$

and

$$\Lambda_n = \int_V \psi_n^2(\boldsymbol{r}_0)dV(\boldsymbol{r}_0) \tag{43}$$

Expression (42) shows that modes amplitude is affected by unsteady heat release rate perturbations and the boundaries of the system. When combustion is not affected by p' fluctuations, this corresponds to weakly coupled interactions, while strongly coupled

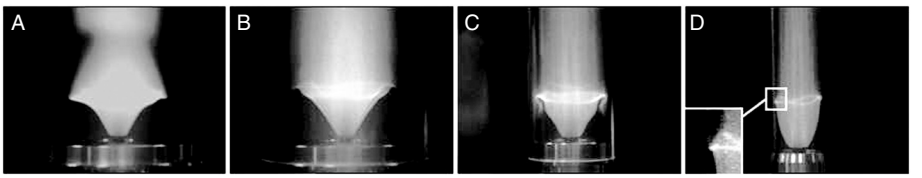


Figure 13: Views of the steady inverted conical flames, with different confinement ratios. $\overline{v} = 2.7$ m s^{-1}, $\Phi = 1.03$ (cases A, B, C). Case D picture is obtained for different flow conditions: $\overline{v} = 1.62$ m s^{-1}, $\Phi = 0.96$.

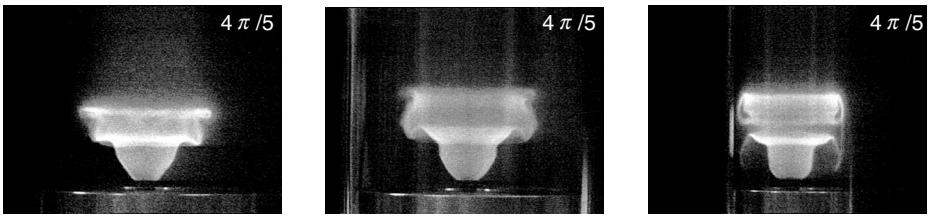


Figure 14: Flame motion during self sustained instabilities. Left: unconfined case $\xi = 0$, center: $\xi = 0.32$, right: $\xi = 0.64$.

interactions occur when the unsteady combustion also depends on p' fluctuations. The presence of boundaries play a significant role. They have two main effects. They modify the reflection response of the system, but they also modify the flame dynamics. The second item is less well documented but is clearly important. The presence of boundaries not only modifies the structure of the mean flow but also influences the flame dynamics. This is demonstrated in a set of recent experiments in which the lateral confinement is varied systematically (Birbaud et al. (2007)). The confinement factor $\xi = d/D$ is defined as the ratio of the injector diameter to the flame tube diameter. Changes in ξ modify the steady flame structure as can be seen in figure 13.

The influence of confinement on the flame motion is examined by measuring the response for different diameters of the duct bounding the flame. It is found that when the confinement ratio is low, the time delay between the velocity signal at the burner outlet and heat release perturbations remains essentially constant indicating that the flame is weakly affected by the presence of the wall. As the confinement ratio is increased, the time delay and the gain of the flame transfer function are modified. The interaction with the wall changes the dynamics of the flame tip, the flame wrinkling geometry in the vicinity of the boundary. This is illustrated in figure 14 which shows three direct views of the flame during self-sustained oscillations.

The amplitude and fine structure of the radiated sound field is also changed as can be seen in figure 15 which compares time records obtained in an open flame case ($\xi = 0$) and in a confined case ($\xi = 0.64$).

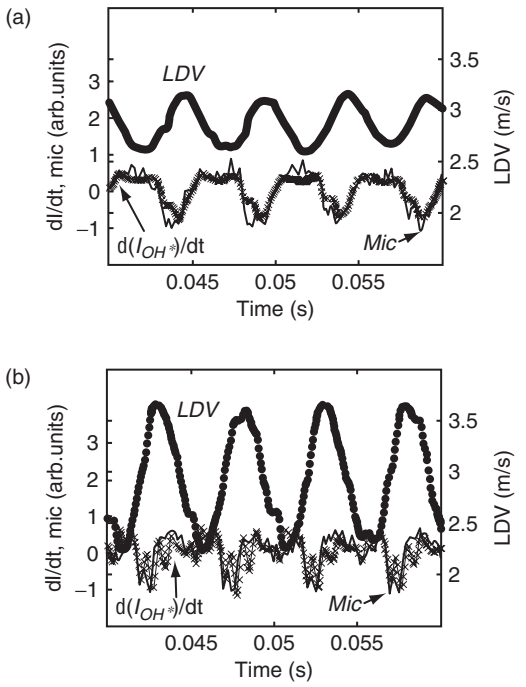


Figure 15: Signals recorded during self sustained instabilities. Short burner (BC). (a) no tube $\xi = 0$, (b) $l = 100$ mm (TC) $\xi = 0.64$. • velocity signal (m s^{-1}) measured in the injection plane, × rate of change of light intensity, solid line: signal recorded by microphone M_2.

It is interesting to note that in both cases the sound field and the time derivative of the light intensity radiated by the flame are in good agreement except for the highest frequencies.

To summerize, three types of configurations can be identified where sound sources and waves may interact. The first one is a freely propagating flame, the second is a flame stabilized on a burner in an unconfined environment and the last one is a flame enclosed in a combustor. These situations are schematically represented in figure 16 together with a representation of the corresponding acoustic boundary conditions of the system.

The only configuration where sound reflection is absent corresponds to freely propagating flames schematically presented in figure 16a (Thomas and Williams (1966); Belliard (1997)). In the second configuration (figure 16b), acoustic waves reflected back towards the burner may interact with the burner acoustics used to stabilize the flame. This kind of setup was used in section 4 to study the flame response and the noise radiated by flames submitted to incoming flow perturbations. In the absence of forcing, it is found that these systems exhibit a wide variety of unstable regimes. The origin of the coupling mechanism is always associated with combustion noise radiated

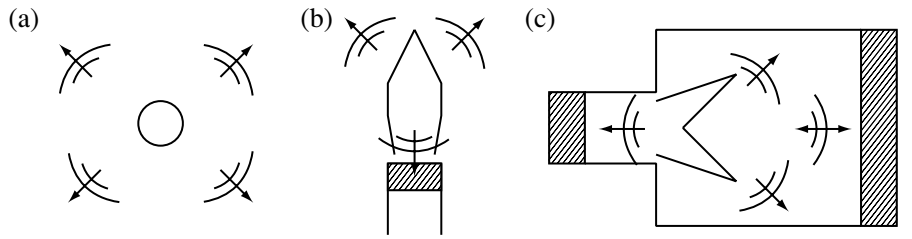


Figure 16: (a) freely propagating flame. (b) unconfined flame stabilized on a burner. (c) flame enclosed in a combustor.

back towards the burner and triggering the instability (Schuller et al. (2003); Durox et al. (2005); Noiray et al. (2006)). In forced flow configurations, these experiments belong to weakly coupled interactions as defined above because the structure of the unsteady flow at the burner outlet is primarily determined by the flow forcing device that imposes the external modulation. The flow in this region remains thus mostly unaffected by the acoustic field radiated by the flame back towards the burner. But in the absence of forcing, strongly coupled interactions observed between the flame and the burner acoustics underline the determinant role of the outlet impedance of the burner in the system stability (Schuller et al. (2003); Noiray et al. (2006)). This impedance viewed by the flame in the upstream region should also affect combustion noise generation from unconfined flames stabilized on a burner. While this discussion is based on laminar flame experiments under self sustained oscillations, similar conclusions should also hold for noise production from unconfined turbulent flames stabilized on a burner, but this point is apparently not documented.

The last configuration presented in figure 16c is perhaps the most complicated and the less well understood. A flame stabilized inside a combustor generates sound waves which interact with all the boundaries of the system but also may interact with the mean flow flow inside the combustor. While the walls essentially reflect acoustic waves, the flame dynamics can also be modified by the presence of the walls (Kendrick et al. (1996)) and may affect the burner stability. In general, the flames are stabilized away from the combustor walls and are more sensitive to modifications of the termination impedances of the system. As in unconfined configurations, the flame dynamics is sensitive to the unsteady flow at the inlets of the system. This is exemplified by Paschereit et al. (2000); Balachandran et al. (2005) in forced flow configurations or by Tran et al. (2007) in a study of the modifications induced on the combustion stability when the inlet reflection coefficient is changed. The conclusion is that the flame dynamics and thus the combustion noise sources are particularly sensitive to flow and acoustic perturbations in the upstream region where the flame is stabilized (Yu et al. (1991)). Moreover in a confined environment, combustion may also be sensitive to modifications of the flow in the downstream region of the flame in response to fluctuations of the flow field at the combustion chamber outlet. Effects of modifications of the outlet reflection

coefficient on the combustor stability and the flame dynamics was studied for example by Paschereit et al. (2000); Eckstein et al. (2006). Different structures of unstable modes were observed depending on the modulus of the downstream reflection coefficient. The mean flow inside the combustor plays also a significant role in sound production. The last mechanism which may affect flame acoustics coupling is due to entropy fluctuations. The flow in the burnt gases of combustors operating at high pressures is often submitted to a strong pressure gradient as it is generally terminated by the bladerows of a turbine or by a nozzle. Hot spots convected from the reaction zone and accelerated in this gradient induce pressure perturbations propagating upstream and downstream from the gradient region. The pressure perturbation propagating in the downstream direction constitutes a supplemental source of noise named "indirect noise" or "entropy noise" (Bake et al. (2007)). The mechanism is well demonstrated in the case of a compact chocked nozzle which generates sound in response to incident entropy waves (Marble and Candel (1978)). The following boundary condition is often used to describe this mechanism at the outlet of a combustor:

$$\frac{p'}{\bar{p}} = 2\frac{u'}{\bar{u}} + \frac{\rho'}{\bar{\rho}} \tag{44}$$

In this expression ρ' are associated with density fluctuations due to incoming entropy waves. Effects of these entropy waves on the combustor stability were studied by many authors with diverse conclusions (Heitor et al. (1984); MacQuisten and Dowling (1993); Eckstein and Sattelmayer (2006)). It is generally admitted that convection of entropy fluctuations may play a significant role in the development of instabilities if the flame region is not too far away from the nozzle exit, but this will depend on the dispersion and dissipation of these waves as they travel through the system. Like the question on the relative importance of indirect to direct noise contribution on the overall sound radiation from a combustor, the relative importance of entropy modes in the system stability also depends on the specific configuration under investigation and is quite sensitive to the combustor geometry, operating conditions and to the complex unsteady reactive flow.

7. COMPUTATIONAL COMBUSTION ACOUSTICS

Much of the current effort in the field of combustion noise is concerned with the development of computational tools for the estimation of the noise radiated by flames. Computational combustion acoustics (CCA) is in a naissant state and it borrows many of its tools and concepts from two more mature domains: (1) Computational aeroacoustics (CAA, see Mankbadi (1999); Tam (2006)) and (2) Computational Flame Dynamics (CFD, see Candel et al. (2007)). The prediction of noise radiated by combustion raises two major issues:

(1) It is first important to determine the noise sources in the flame region. As clearly apparent from the previous sections this requires a precise modeling of the flame dynamical properties. The determination of the noise sources has to be carried out in a region of finite size which encloses the flame.

(2) The second problem is that of sound propagation from the flame zone to the farfield. As in CAA, the dimensions of the propagation domain are usually much larger than those of the flame region (typically 50 to 200 D where D is the characteristic size of the combustion region).

The computational domain sizes corresponding to these two problems are widely different underlining the multiscale nature of the problem. Tackling both issues in the same modeling framework may not be the best solution. There are cases in nonreactive flows where this has been attempted (see for example recent articles by Birbaud et al. (2006) on modulated turbulent jets and by Bogey and Bailly (2006) on jet noise radiation).

In combustion noise most of the current investigations deal with the two issues separately.

7.1. Combustion noise source modeling

Methods reviewed in this section are devoted to the determination of noise sources inside and in the near vicinity of the combustion process. The simulation domain which is envisaged has typically the size of the combustion chamber or of a few injector diameters in the case of jet flames.

As in CAA, one can distinguish three general classes of methods for estimating noise emission from turbulent flames. These classes have been mainly explored in aerodynamic noise investigations pertaining to practical situations in ground and air transportation and in industrial flow noise problems (Ozyörük and Long (1996); Tam (2006)).

- Stochastic methods rely on a synthetic field of fluctuations to represent the instantaneous distribution of sources in the flowfield. The method has been explored in CAA (Bechara et al. (1994); Bogey et al. (2002)). The method is attractive because it does not require costly simulations and because it is applicable to complex geometries. It is however not so easy to construct a suitable field of fluctuations having the properties of real eddies in a combustion region. This is probably why the stochastic method is apparently not considered in the current CCA effort.
- Statistical methods are based on elaborate source models. The source characteristic parameters like time and spatial scales, correlation function and source intensity are usually deduced from a steady or unsteady state calculation of the flame based on average Navier-Stokes equations. This scheme has been extensively explored in CAA most notably in the case of free jets at subsonic or supersonic conditions (Bailly et al. (1994, 1996, 1997), Bechara et al. (1995), Tam and Auriault (1999)). Due to the reasonable computational cost associated with RANS and $k - \epsilon$ simulations, this class of methods is now widely used in design applications mostly for non-reactive situations. Statistical source modeling is not as advanced in the case of combustion noise.
- Direct Numerical Simulation (DNS) and Large Eddy Simulation (LES) constitute the most novel methods of combustion source determination. In both cases, the simulation provides instantaneous distributions of fluctuations in the flowfield, in a region which includes the flame. In reactive DNS, investigations were initially carried out in two dimensions and for flames with simple chemistry. Three-dimensional

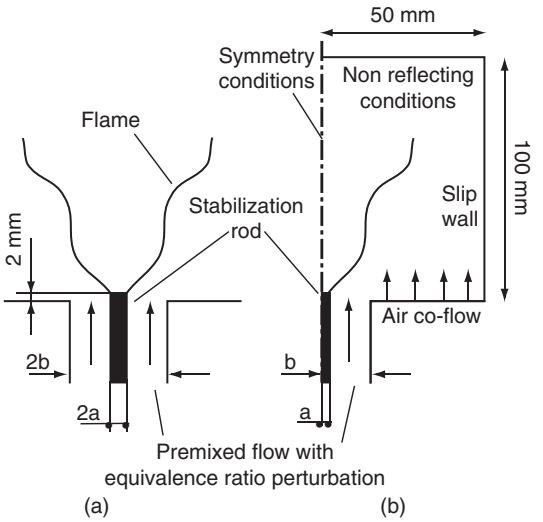


Figure 17: Computational domain and boundary conditions. The flame front position is drawn as a solid line.

direct simulations have progressively taken over and reasonably complex chemistry has been included in some of these calculations. DNS has advanced the state of the art in turbulent combustion modeling by providing a detailed view of the dynamical behavior of flames in a turbulent environment (Poinsot et al. (1996)). For laminar cases, the improvement of computer capacities has led to new possibilities. For simple geometries, it is now possible to simulate the flame dynamics in different coupling situations. DNS thus constitutes a useful tool in the identification of noise sources in simple laminar configurations.

Most of the following discussion will focus on the third class of methods. It is first interesting to illustrate the possibilities of DNS by considering the response of a laminar "V"-flame to an equivalence ratio modulation. It is known from previous studies and described in what follows that the heat release rate can be strongly influenced by the incoming modulation, leading to noise radiation.

7.2. DNS of a laminar "V"-flame submitted to equivalence ration modulation

These calculations are carried out in the configuration sketched in figure 17 of an inverted dihedral ("V") flame stabilized on a central rod. The rod size is $2a = 3$ mm and the injector lip lateral position $b = 5.5$ mm. Premixed reactants are injected at the bottom of a 10 mm long channel. The width of this channel is $d = b - a = 4$ mm. The other dimensions are gathered in figure 17. One assumes that the burner is long enough in the third direction, so that 2D simulations can be carried out as a first step. In a strictly laminar situation, the problem is symmetrical, and it is only necessary to simulate one

half of the domain. The following calculations are carried out with the AVBP code developed at CERFACS and IFP. AVBP is dedicated to simulate turbulent compressible flames using Large Eddy Simulations (LES) on structured or unstructured meshes, but it can also be used for DNS in simple laminar situations. Integration schemes ensure a third order precision in space and time (Colin and Rudgyard (2000)). The steady state flame is obtained using a Lax-Wendroff second order scheme in space and time. Calculations are then pursued with a numerical scheme designated as TTGC which provides third order accuracy.

An unstructured grid is used to mesh the domain with a high level of refinement in the region where the flame is expected to develop. The mesh is densified near the steady flame solution. The angular location α of the "V" flame can be deduced from simple calculations, which take into account the imposed flow U, the laminar burning velocity S_L and the deviation angle β of the streamlines originating from the fresh mixture. The flame front is determined in these calculations using an Arrhenius chemistry model without any artificial thickening of the front, as would be necessary in a LES framework. This is achieved by making sure that the reaction rate is resolved with at least 10 grid points. This very high level of refinement is conserved in the whole region situated on the two sides of the steady flame because it is required to allow a correct calculation of the modulated flame.

A detailed kinetic scheme is not needed here and one may use a single step reaction providing accurate values of S_L in the equivalence ratio range of interest:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O \tag{45}$$

and the corresponding volumetric rate of reaction $\dot{\omega}$, given in mol m^{-3} s^{-1} is defined by an Arrhenius law:

$$\dot{\omega} = A\left(\frac{\rho Y_{CH_4}}{W_{CH_4}}\right)^{\nu_{CH_4}}\left(\frac{\rho Y_{O2}}{W_{O2}}\right)^{\nu_{O2}} \exp\left(\frac{-E_a}{RT}\right) \tag{46}$$

with a pre-exponential factor $A = 1.1\ 10^{10}$ S.I., an activation energy $E_a = 20$ kJ mol^{-1} and exponents $\nu_{CH_4} = 1$ and $\nu_{O_2} = 0.5$. These values were adjusted to yield the correct laminar burning velocities. The computational results displayed in Figures 18 and 19 show the effect of equivalence ratio modulation on the flame wrinkling and on the heat release rate. This last quantity is nearly periodic because the initial transient is not yet evacuated and the system has not quite reached a steady cyclic state. The signal features a slow growth followed by a sudden drop. The shape of the radiated sound field can be easily deduced by taking the derivative of the global heat release signal and making use of expression (2).

The previous example serves to show that the dynamics of flames can be simulated with DNS and that results of calculations can be used to extract noise radiation from the flame. This is however limited to rather simple geometries, low Reynolds Numbers and moderate Damköhler numbers. These limitations can be overcome with LES, in which

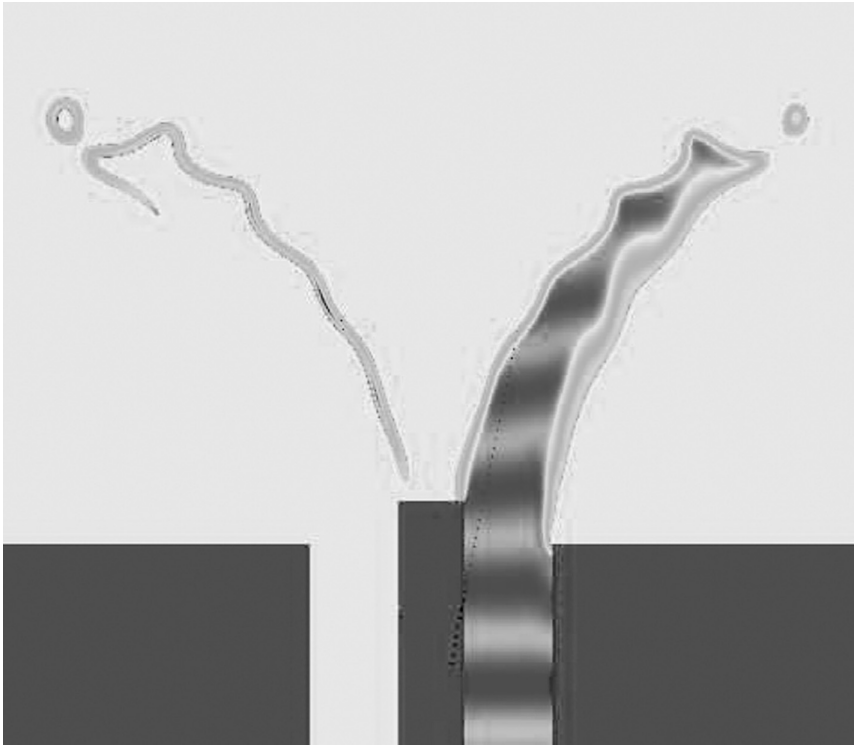


Figure 18: Two-dimensional "V"-flame submitted to an equivalence ratio modulation. Frequency of the CH_4 mass fraction modulation: 575 Hz. Amplitude of modulation: 0.1 Left: reaction rate. Right: Y_{CH_4}.

the larger structures of the flow are calculated, while only the subgrid scale structures are modelled. LES has been widely used in two fields, which are closely related to combustion noise: Computational Combustion Dynamics (CCD) for combustion instability analysis, and Computational AeroAcoustics (CAA). Progress in these two fields is briefly reviewed in what follows. Advances in Computational Combustion Dynamics have been quite substantial. This research was mainly motivated by the difficult problems arising in modern combustion technology and most notably by combustion instabilities affecting premixed gas turbine combustors and many other high performance devices like liquid rocket motors. Instabilities result from the strong coupling between acoustics and combustion. The general objective has been to develop numerical tools allowing an analysis of mechanisms leading to instabilities and eventually to derive predictive tools which could be used in engineering design. There are three central problems in CCD:

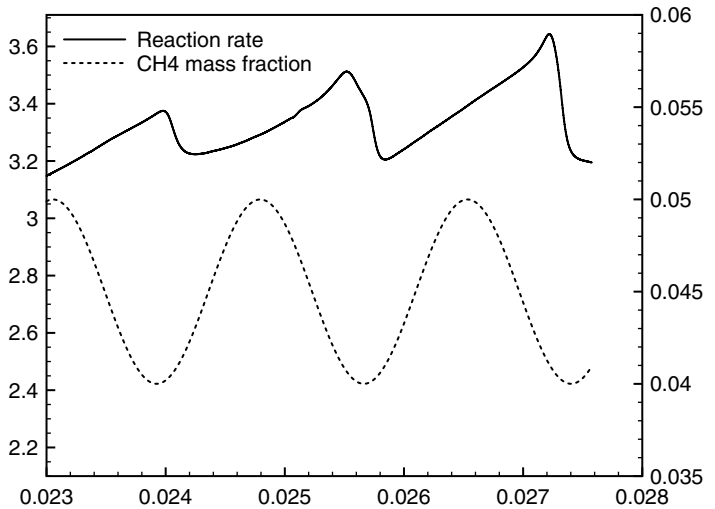


Figure 19: Temporal evolution of the CH_4 mass fraction and the reaction rate during a few cycles of modulation. Periodicity has not been fully obtained yet.

- The combustion model has to account for the large scale dynamics of the reactive system and for the subgrid scale effects. Many of the models developed for RANS can be adapted to LES (see Veynante and Vervisch (2002); Poinsot and Veynante (2005) for reviews). The combustion model must describe the flame dynamics and its coupling with acoustic waves. This last capability can be tested numerically to see if the model correctly retrieves the acoustic-flame interactions, in simple laminar situations (Ducruix and Candel (2004); Birbaud et al. (2007)) or more complex practical turbulent configurations (Roux et al. (2005); Selle et al. (2006). One may infer from these simulations that the production of sound by flames can b e suitably described at least in the premixed regime where noise production is associated with flame surface fluctuations but this still requires further validations.
- A numerical scheme that is neither too dissipative nor dispersive, so that acoustic propagation can be suitably described in the flame nearfield allowing a suitable coupling between the sound pressure field and heat release fluctuations. One such scheme is due to Colin and Rudgyard (2000) but there are many other options. This point is even more crucial in the CCA framework if one wishes to describe the combustion source field and its radiated sound with the same numerical scheme.
- Adapted conditions, for inlets, outlets and free field conditions. This issue has been studied extensively (see Poinsot and Veynante (2005) for a review and Selle et al. (2004) for recent work on non-reflecting boundary conditions). Specific developments have also been proposed for acoustic modulation at a boundary (Kaufmann et al. (2002); Ducruix and Candel (2004)) and Candel et al. (2007) for time domain impedance conditions.

There are many parallel developments in the CAA domain and the literature is quite substantial (see the following publications for sample simulations of subsonic and supersonic jet noise: Bastin (1996); Freund (2001); Zhao et al. (2001); Bogey and Bailly (2006)). Among other important issues, work has focused on the following items:

(1) Non dissipative and non dispersive schemes. This issue is central to CAA and many publications describe developments of highly accurate schemes (see for example Tam and Webb (1993); Nance et al. (1997); Rizzetta et al. (2003); Bogey and Bailly (2004)).
(2) Adapted conditions for inlet and outlet conditions (see Colonius et al. (1993); Tam et al. (1998)).
(3) Subgrid scale modeling (for example Seror et al. (2000)).

Clearly much can be shared between CAA and CCD but there are also some significant differences. In combustion, it is important to suitably represent the flame dynamics and the associated temperature and density gradients and these may not be so easy to accomodate within the CAA methodologies. Another distinguishing feature is that combustion takes place in complex geometries which are better handled on unstructured meshes.

7.3. LES of turbulent conical flame

After these general considerations it is worth using the simple case of a turbulent conical flame to illustrate the application of LES to combustion noise. The following calculations are carried out with the AVBP code (see section 7.2). LES is well adapted to the problem at hand because most of the heat release perturbations which are at the origin of the radiated sound are induced by the large scale flame dynamics (the central frequency of the sound emitted is typically given by $f_c \simeq U/L_f$ see Rajaram et al. (2006)). The large eddies carry the major part of the turbulent kinetic energy and one may reasonably assume that, the smaller scale eddies will have a weaker influence and that their dynamics can be described by the subgrid scale models used in standard LES calculations.

In LES the flow variables are spatially filtered:

$$\overline{f(\boldsymbol{x},t)} = \int_{-\infty}^{\infty} f(\boldsymbol{x}', t) G(\boldsymbol{x}' - x) \boldsymbol{dx}' \tag{47}$$

where G is a localised filter and $\overline{f(x, t)}$ is the filtered value of the variable $f(\boldsymbol{x}, t)$. In the description of turbulent multispecies flows, the primary variables are the velocity vector $v_i(\boldsymbol{x}, t)$, the species mass fractions $Y_k(\boldsymbol{x}, t)$ and the associated quantities, the total energy E and the density $\rho(\boldsymbol{x}, t)$, where $E(\boldsymbol{x}, t) = e + 1/2 v_i v_i$ and $e(\boldsymbol{x}, t) = \sum_{k=1}^{N} e_k Y_k(x, t)$ (e being the mixture sensible energy). It is assumed that all constituents follow ideal gas laws, and that the viscous stress tensor, heat flux and species transport follow standard gradient laws. Under the assumption that diffusion coefficients are not influenced by the unresolved fluctuations, molecular transport terms become a function of the filtered field (Poinsot and Veynante (2005)). The Navier-Stokes equations are then written in terms of filtered quantities, and are solved in conjunction with a subgrid-scale closure

(SGS). In compressible flow situations, it is convenient to weight the filter by the density and use Favre filtered quantities like υ_i for the velocity vector and $\widetilde{S}_{ij}$ for the resolved strain rate tensor. In the present illustrative context, it is also convenient to use a standard Smagorinsky model to express the subgrid scale stress tensor It is known from previous studies that the Smagorinsky model is not the best subgrid scale closure because of its inherent dissipative nature. The reasons underlying the present choice are however that (1) This scheme is easiest to implement on an unstructured grid, (2) The initial developing part in the flame is finely meshed to minimize the contribution of the unresolved scales (3) The flow response is tightly coupled to the large scale dynamics, which are well described in the LES framework and only weakly influenced by the SGS model. It is reasonable to consider that the model will yield adequate predictions of energy-containing components in high Reynolds number flows, provided that the filter size Δ is well within the inertial subrange.

The same single step global chemical reaction as in the laminar case is considered to proceed to completion generating water vapor and carbon dioxide: and the volumetric rate of reaction $\dot{\omega}$, given in mol m^{-3} s^{-1} is defined by the same Arrhenius law (46).

One of the difficulties of combustion LES is that the thickness δ_L^0 of a premixed flame is about 0.1 to 1 mm and is generally much smaller than the LES mesh size. This difficulty can be overcome in various ways (Poinsot and Veynante (2005)). One convenient method consists in thickening the flame artificially, a method devised by O'Rourke and Bracco (1979) in a different context and exploited by Thibaut and Candel (1998) to investigate flashback dynamics. In the LES context, the Thickened Flame model (Colin et al. (2000)) provides a suitable description of the flame dynamics and subgrid effects on combustion. In TFLES, the flame is artificially thickened by a factor F to resolve its inner structure on the numerical mesh (in general, combustion is a subgrid scale phenomenon as flames are thinner than typical filter sizes (Poinsot and Veynante (2005))), keeping constant the laminar flame speed. However since the flame is made thicker it is also less sensitive to small scale turbulence. This can be compensated by an efficiency function E which measures the wrinkling of the flame front at the sub-grid scale level.

In practice, the subgrid scale diffusive species flux vector $\overline{J_{i,k}}^t$ is replaced by $EF\overline{J_{i,k}}^t$, the subgrid scale heat flux vector $\overline{q_i}^t$ is replaced by $EF\,\overline{q_i}^t$ and the filtred source terms $\overline{\mathbf{s}}$ becomes:

$$\overline{\mathbf{s}} = \left(0,0,0,\frac{E}{F}\overline{\dot{\omega}}_T + Q_r\left(\tilde{Y}_K,\tilde{T}\right), -\frac{E}{F}\overline{\dot{\omega}}_K\right)^T \tag{48}$$

where $\overline{\dot{\omega}}_k$ is computed from the filtered temperature and mass fractions, using Arrhenius laws developed for laminar flames, $\overline{\dot{\omega}}_k = \dot{\omega}_k(\widetilde{Y}_k, \widetilde{T})$. The total energy is the sum of filtered heat release and radiative power, $Q_r(Y_k, T)$ that is computed using the filtered temperature and mass fractiones.

As shown in Colin et al. (2000), this formulation propagates a premixed flame front of thickness $F\,\delta_1^0$ at the subgrid scale turbulent flame speed $E\,s_L^0$ where s_L^0 and δ_L^0 denote respectively laminar flame speed and laminar flame thickness. The efficiency factor is

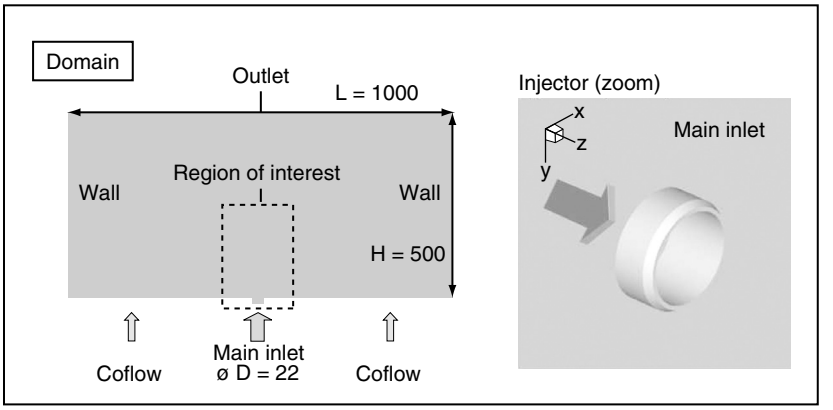


Figure 20: Schematic view of the 3D computational domain, and boundary conditions adopted for the calculations. Right side, close-up view of the injector geometry. Dimensions are given in mm.

expressed from turbulence and flame characteristics and the thickening factor according to Colin et al. (2000):

$$E = \frac{1+\alpha\Gamma\left(\frac{\Delta e}{\delta_L^0}, \frac{u'_{\Delta e}}{s_L^0}\right)\frac{u'_{\Delta e}}{s_L^0}}{1+\alpha\Gamma\left(\frac{\Delta e}{F\delta_L^0}, \frac{u'_{\Delta e}}{s_L^0}\right)\frac{u'_{\Delta e}}{s_L^0}} \tag{49}$$

where the function $\Gamma(\Delta_e/\delta_L, u'_{\Delta e}/s_L)$ measures the ability of a sub-grid scale turbulence of intensity $u'_{\Delta e}$ (filter size Δ_e) to wrinkle a premixed flame front of thermal thickness δ_L propagating at the laminar burning velocity s_L (see Colin et al. (2000) for details). In this expression α is a model parameter to be specified. One of the advantages of the TFLES model is that it uses Arrhenius laws and can therefore handle, at least qualitatively, thermal ignition, heat losses or quenching processes.

The computational flow configuration is designed to suitably deal with the combustion process and allow the outwards propagation of acoustic waves. The geometry represented schematically in Fig. 20 idealizes that used in unconfined premixed turbulent flame experiments. The jet is exhausted from a cylindrical nozzle (D = 22 mm) placed on the lower side of a large 3D volume. The injection section comprises a 15 mm cylindrical element to establish an initial velocity profile, bounded by a lateral wall of 2 mm thickness (see enlargement in Fig. 20). The nozzle features a 1 mm edge which is finely meshed. The lateral boundary is located at 22 D from the centerline and the outlet is located at the same distance from the nozzle exhaust. A region of interest is defined, surrounding the jet, featuring a 30 mm radius and 130 mm height. Beyond this zone, the mesh is progressively coarser. An absorption region is

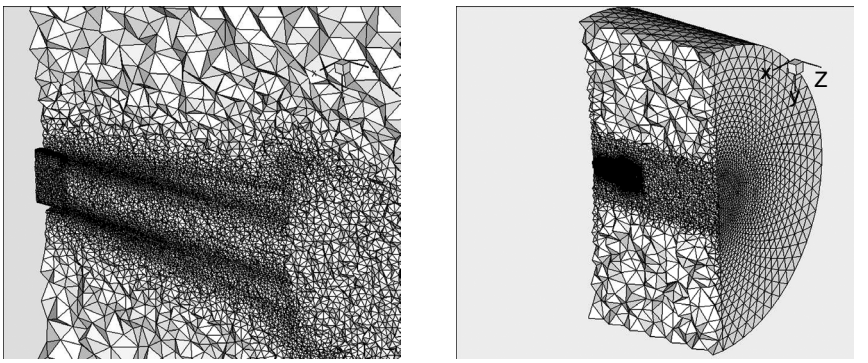


Figure 21: Views of the mesh adopted for the simulations. Left side: close-up view of the region of interest. Right side: global view of the computational domain.

defined in the vicinity of the lateral boundaries to avoid acoustic wave reflection. Slip conditions are used on these boundaries. Navier-Stokes characteristic conditions employed on the inlet allow a proper control of acoustics at that boundary. The non-reflecting condition applied at the outlet includes a relaxation factor which keeps the mean pressure around a predetermined ambient value (Poinsot and Veynante (2005)). The fully unstructured mesh is composed of 2,670,000 tetrahedra (455,000 nodes). Mesh refinement is enforced in the region of interest which comprises the main velocity gradient regions and corresponding turbulent zones in the high velocity case. Typical views of the mesh are displayed in Fig. 21.

A typical result of calculation is shown in figure 22. A stoichiometric mixture of air and methane is injected at a mean velocity $U = 30$ m s^{-1} and a temperature $T = 300$ K. The fuel mass fraction field on the left shows an instantaneous flame configuration while the corresponding iso-surface of heat release displayed on the right shows local spatial fluctuations. This iso-surface is superimposed on the temperature field distribution in the *xy* plane. The calculation provides the heat release distribution which can be used to determine the global rate of heat release in the flame and calculate the sound field radiated by making use of equation (2). It is also possible to use the spatial distribution of heat release as a source term in a perturbed set of Euler equations and determine the resulting sound field. This second step is examined in what follows.

7.4. Simulation of acoustic propagation

It has been shown that sources of combustion noise can be determined by making use of current combustion LES tools. It remains to define methods for the propagation of sound to the farfield. In contrast with combustion instability problems, where acoustic propagation takes place in a finite size internal flow such as a gas turbine the sound propagation has to be simulated over a large scale domain which covers at least a few wavelength. A direct method consisting in calculating the reactive flow and the radiated sound field at ounce would require higher order discretization schemes and clean nonreflecting boundary conditions.

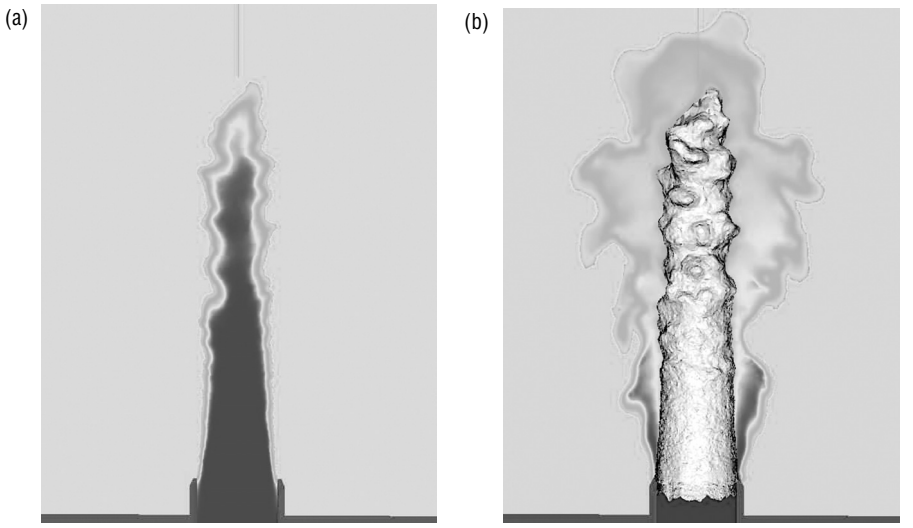


Figure 22: Conical turbulent flame. (a) Instantaneous CH4 mass fraction field. (b) Reaction rate iso-surface superimposed on the temperature field in the *xy* plane.

Indirect or hybrid schemes use the fluctuations obtained from the large eddy simulations to calculate in a second step the sound radiation by making use of perturbed Euler equations including noise source terms (Bogey et al. (2002)). It is also possible to determine the farfield by using for example integral expressions of the farfield sound intensity or by extrapolating nearfield values determined on a surface enclosing the source region, (Kirchhoff surface integration, see Gamet and Estivalezes (1998); Andersson et al. (2005), Rahier et al. (2004)). Another method consists in first determining the space time spectral density of the sources and exploiting integral relations in the frequency domain.

Hybrid schemes are explored for example by Flemming et al. (2007) who use three dimensional LES to calculate the flowfield of a nonpremixed turbulent jet flame and a two-dimensional wave equation to determine the sound field radiated by the flame. The starting point is Lighthill's equation cast in the form

$$\frac{\partial^2 p'}{\partial t^2} - c^2 \frac{\partial^2 p'}{\partial x_i \partial x_j} = c^2 \frac{\partial^2}{\partial x_i \partial x_j}\left(\rho \upsilon_i \upsilon_j - \tau_{ij}\right) + c^2 \frac{\partial^2}{\partial t^2}\left(\frac{1}{c^2} p - \rho\right) \tag{50}$$

It is then argued that the fluctuating density exceeds the other three terms in the right hand side by at least two orders of magnitude and the previous equation is replaced by a two-dimensional version:

$$\frac{\partial^2 p'}{\partial t^2} - c^2 \left(\frac{\partial^2 p'}{\partial x^2} + \frac{\partial^2 p'}{\partial y^2}\right) = -c^2 \frac{\partial^2 \rho}{\partial t^2} \tag{51}$$

The source term $-c^2\partial^2\rho/\partial t^2$ is obtained from a low Mach number large eddy simulation.

One critical point in the previous hybrid scheme and more generally in methods which couple LES and a wave propagation solver is the identification of the source term. The choice has a profound influence on the result. It is important to make sure that the term selected to represent the source is indeed adequate. A comparison of equations (20) and (51) indicates that this will be so if

$$(\gamma-1)\dot{Q}' \simeq -c^2\frac{\partial\rho}{\partial t} \tag{52}$$

It is clear that the extraction of the source terms to be included in the propagation code will determine the quality of the sound field radiated by the combustion process. This aspect is discussed for example by Bui et al. (2007).

8. CONCLUSIONS

While combustion noise has received less attention than jet noise or rotating machinery noise, progress has been accomplished in many respects. Sources of indirect combustion noise identified in the early 70's to result from coupling between convected inhomogeneities and mean pressure gardients are now the subject of further evaluation based on modern computational tools. Direct combustion noise sources also identified during the same period have been studied in a series of detailed investigations relying on perturbed flame configurations. The basic expressions providing the sound field in terms of heat release fluctuations or flame surface area are now well validated in the premixed flame case. These experiment have also provide a link between flame dynamics and noise radiation. While the early correlations of the sound output of turbulent flames were relatively imprecise some more recent efforts have provided more adequate results at least for premixed turbulent jet flames. Much of the recent effort has concerned the development of computational tools for combustion noise estimation. Initial results of this effort are promising but this particular field of research is clearly less advanced than that of computational aeroacoustics. One of the future challenges will be to devise reliable computational tools which would be capable of reproducing well controlled experiments on perturbed laminar flames. Another objective will be to develop the suitable code coupling methodology which will provide noise radiation estimates and suitable spectral densities.

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